

MATH SURVEY



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NOTICE TO AUTHORS

MATH SURVEY is the annual student journal of the Stuyvesant High School Mathematics Department. Students are encouraged to submit papers written during research courses or programs, as well as synopses of subjects learned in school or in private study. Stuyvesant faculty are occasionally invited to submit material with the understanding that MATH SURVEY is primarily a vehicle for publishing student work. Written work need not represent original theory, but may instead summarize or explain a subject. Appropriate subjects include all aspects of mathematics and computer science not encountered in the standard high school curriculum.

Each paper submitted to MATH SURVEY will be read by a faculty referee, who will convey notice through the editor of whether the paper has been accepted, and if so, what improvements are to be required; these may be mathematical or stylistic. The editorial staff reserves the right to alter final submitted drafts in point of style, wording, and layout.

Our readers expect a high level of both scholarship and exposition. To be a good candidate, an article should first contain important and challenging mathematical ideas. The writing should be clear and straightforward, so that the content may speak for itself. Sufficient background should be given so that a non-expert may learn and profit from each article.

Authors should submit their work in the L^AT_EX markup language if possible. Other submissions will be rendered in L^AT_EX at the discretion of the editors. To learn our formatting style, as well as how to properly cite other published materials, please examine the present volume.

Submit work to mathsurvey@gmail.com.

EDITOR'S NOTES

ADAM SEALFON

Dear Reader,

The beauty and variety of mathematics is truly dazzling. Stuyvesant High School has a long tradition of excellence in mathematics. MATH SURVEY has been the official publication of the Mathematics Department since 1927. We are proud to be contributing to its continuity.

The work of many individuals made this publication possible. Thanks to our contributors for their submissions and for their love of mathematics. I am very grateful to the staff of MATH SURVEY for the tireless work they have invested in this journal. I would particularly like to thank the Managing Editor, Muzhou Lu, and the Associate Editors, Milo Beckman, Lawrence Chan, Benjy Hirsch, Yakov Pechersky, Taylor Sutton, Yang Tang, and Kenny Yu, without whom this would not have been possible.

I would especially like to thank our faculty advisers, Mr Joseph Stern and Mr Ashvin Jaishankar, for all their support, help, advice and expertise. The guidance they provided was instrumental in bringing this volume to print.

Finally, we are honored to have had the support and confidence of Mr Teitel, Ms Ferrara, and the entire Mathematics faculty.

Hope you enjoy!

Sincerely,
Adam Sealfon
Editor

FOREWORD

PETER B. SHALEN, PhD[†]

Peter Shalen is a professor in the Department of Mathematics, Statistics and Computer Science at the University of Illinois, Chicago. A native of New York City, he graduated from Stuyvesant High School in 1962 and Harvard College in 1966, prior to earning a PhD from Harvard University in 1972. Professor Shalen is widely recognized as a leading researcher in the field of low-dimensional topology. He has held visiting professorships at such distinguished universities as the University of Chicago, the University of Paris, Columbia University, and the Courant Institute at NYU. Professor Shalen and his wife Catherine reside in Chicago, where they enjoy the company of their four young grandchildren.

I graduated from Stuyvesant in 1962 and was a member of the math team.

When you think of the 1960's you probably think of college students rebelling against authority. Well, my friends and I, like many Stuyvesant students, were precocious and were math geeks (even if the term was unknown back then). So we did our rebelling while still in high school, and we did it by arguing with our teachers about what real mathematics was, and how it should be taught.

One focus of our rebellion was the math squad meetings that were held early in the morning, during what I think was called 'zero period'. At one point my friend Ackerman was captain of the math team and I was co-captain. We were supposed to drill the other students on the squad about solving problems of the kind that might appear at math meets. Instead, we decided to spend the time lecturing on how to deduce basic properties of the natural numbers from the Peano axioms.

After a couple of sessions of this, Mr. Schwartz, who was coach to the math team, explained to us that although he enjoyed this kind of thing, it had nothing to do with what the squad was supposed to be doing. So we started holding our lectures in a park across the street from the school — even earlier in the morning, before the math squad met ('minus one period', I guess). This went on for several weeks, I believe, with a small cluster of diehards coming to listen to us hold forth. We all wore our coats because the weather was chilly. How we managed without a blackboard I don't clearly remember.

[†]The staff of MATH SURVEY are grateful to Professor Shalen for generously agreeing to share his recollections of life and mathematics at Stuyvesant High School.

Anyway, the down side of this was that I didn't get very good at solving problems of the kind that were given at math meets, and I got bounced off the team once or twice for various periods. I got pretty good at theoretical math, and when I became a research mathematician I had to catch up on problem-solving skills. Most people do things in the opposite order.

I guess what this illustrates is that there are many roads to becoming a research mathematician. I had decided at some point during my sophomore year at Stuyvesant that I would have to be a pure mathematician because I wasn't fit to do anything else. But plenty of the mathematicians I know started out as perfectly normal people.

I still feel very much connected to Stuyvesant. Just a few days ago I got an e-mail from a mathematician who had done his Ph.D. under my direction about 10 years ago. He grew up in Ohio (not that there's anything wrong with that!) but he knows I'm a Stuyvesant graduate, and he was writing because he had just learned that Paul Cohen, a famous mathematician at Stanford who died recently, had also gone to Stuyvesant (class of 1950). I referred my former student to Art Rothstein's wonderful web page on the Stuyvesant math team: <http://173.8.135.113/MathTeam.html>.

On the web page you'll find a picture of Mr. Schwartz, who was still coaching the math team eight years after I graduated — and I believe he'd been doing it for a couple of decades before I got to Stuyvesant. You'll also find out about many former members of the Stuyvesant math team, all the way back to the class of 1939, and what they have become. Quite a few have become mathematicians, including some very famous ones. Others are doing all kinds of things.

The page gives a terrific sense of the continuity of the whole Stuyvesant enterprise, and sums up a lot of what I love about the school.

Peter Shalen

Chicago, IL

The Catalan Numbers

Adam Sealfon

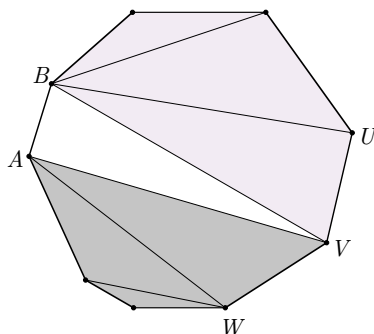
1. INTRODUCTION. The *Catalan numbers* are an important sequence of positive integers that appear in a variety of combinatorial problems. In this article, we will explore a few problems that involve the Catalan numbers and derive two formulas which generate them: one recursively and one explicitly.

2. PROBLEMS AND RECURSIVE FORMULA. We first present four equivalent combinatorial problems. We will demonstrate that the answer to each of these problems is the same sequence of numbers $\{C_n\}$, which satisfies a certain characteristic recurrence relation.

Problem 1. *Compute the number of triangulations of a convex $(n + 2)$ -gon.*

Define C_n to be the number of triangulations in a convex $(n + 2)$ -gon. A triangle has a unique triangulation (itself!) and so $C_1 = 1$. A convex quadrilateral has two possible triangulations, each formed by drawing one of the two possible diagonals between opposite vertices; so $C_2 = 2$. We would now like to find a recursive formula for C_n .

Figure 1: Recursively counting the triangulations of a convex $(n + 3)$ -gon.



Consider a convex $(n + 3)$ -gon $\mathcal{P} = AB \dots UVW \dots$, as shown in Figure 1 above. In each triangulation of $\mathcal{P}$, the edge AB is an edge of exactly one triangle, in this case $\triangle ABV$. This creates two non-overlapping subpolygons, $\mathcal{P}_1 = AVW \dots$ and $\mathcal{P}_2 = BVU \dots$ (colored in different shades). Suppose these have $i + 2$ and $j + 2$ sides, respectively, so that there are $C_i \cdot C_j$ ways to independently triangulate both subpolygons. These triangulations of $\mathcal{P}_1$ and

$\mathcal{P}_2$, together with $\triangle ABV$, form a triangulation of $\mathcal{P}$, and every triangulation of $\mathcal{P}$ involving $\triangle ABV$ arises in this way. Now i and j are not arbitrary, but are instead related in the following way. Removing one of the $i + 2$ sides of $\mathcal{P}_1$ (namely AV), as well as one of the $j + 2$ sides of $\mathcal{P}_2$ (namely BV), and adding back edge AB , we get $\mathcal{P}$. Hence

$$[(i + 2) - 1] + [(j + 2) - 1] + 1 = n + 3, \quad \text{or} \quad i + j = n.$$

Finally, letting V vary through all vertices of $\mathcal{P}$ different from A and B , we find that the total number of triangulations of the $(n + 3)$ -gon $\mathcal{P}$ is given by

$$C_{n+1} = C_0 C_n + C_1 C_{n-1} + \cdots + C_n C_0 = \sum_{i=0}^n C_i C_{n-i}. \quad (1)$$

Here, we have put $C_0 = 1$ by convention (it has to be a convention, since a ‘2-gon’ is a line segment, and we cannot speak of triangulations). The reason for the convention is that when V is adjacent to A (say), then $\mathcal{P}_1$ does not exist, and the number of triangulations of $\mathcal{P}$ involving $\triangle ABV$ is simply the number of triangulations of $\mathcal{P}_2$. The corresponding term of the above sum is just C_n , but to make this fit the general pattern $C_i C_{n-i}$, we must define $C_0 = 1$.

The recursive formula (1) and the initial condition $C_0 = 1$ are sufficient to determine the Catalan numbers. For instance, $C_1 = C_0^2 = 1$, $C_2 = 2C_0 C_1 = 2$, $C_3 = 2C_0 C_2 + C_1^2 = 5$, etc.

Problem 2. *In how many ways can $2n$ people sitting around a circular table shake hands in such a way that each person shakes hands with exactly one other person and no two handshakes cross each other?*

Call the answer to this problem H_n , and let a *handshake configuration* be a set of n pairings of the $2n$ people such that the people in each given pair can shake hands without any of the handshakes crossing. When $n = 1$, we have just two people at the table, and there is clearly only one way to shake hands. So $H_1 = 1$. We now show that H_n satisfies the same recursive formula (1) as C_n . Consider the situation with $2n + 2$ people, in which there are by definition H_{n+1} valid configurations. Look at some particular person, say P , who must shake hands with some person Q . Since no handshake may cross any other, we can compute the number of possible handshake configurations on either side of PQ , and simply multiply them together to get the total number of configurations involving the pairing $\{P, Q\}$.

A slight difficulty is that since H_i is defined as the number of handshake configurations for $2i$ people, we must first verify that the number of people on either side of a particular handshake is even. In fact, after removing P and Q , we have $2n$ people left. These are split into two groups, say of sizes a and b . The a people in one group inherit a valid handshake configuration from the original

configuration presently being separated, since no one from this group can cross PQ to pair up with someone in the other group. But since everyone has exactly one handshake partner, this forces a to be even, and likewise b . Writing $a = 2i$ and $b = 2j$, the number of valid configurations involving the pairing $\{P, Q\}$ is $H_i H_j$. Of course, since $a + b = 2n$, we have $i + j = n$. By letting Q vary through the people besides P , we get one term $H_i H_{n-i}$ for each partition of $2n$ into even parts $\{2i, 2n - 2i\}$. [This accounts for $n + 1$ terms; the remaining $(2n - 1) - (n + 1) = n - 2$ people besides P contribute nothing to the sum, since no valid handshake configuration can involve their being paired with P .] Thus

$$H_{n+1} = \sum_{i=0}^n H_i H_{n-i}.$$

As in the Problem 1, we make the convention that $H_0 = 1$. This is consistent with the value $H_1 = 1$ already computed, so there is no difficulty in starting the sequence at $n = 0$.

Since $\{H_n\}$ and $\{C_n\}$ satisfy the same recursion and the same initial data, it is clear that $H_n \equiv C_n$. It may be rather surprising that this problem turned out to be in some sense equivalent to Problem 1.

Problem 3. *Compute the number of arrangements of n opening parentheses ‘(’ and n closing parentheses ‘)’ in which at least k opening parentheses appear before the k^{th} closing parenthesis for each $k \in \{1, 2, \dots, n\}$.*

The restriction on opening and closing parentheses in this problem has an important rationale having to do with associativity. Often we have some sort of binary product operation satisfying an associative law $(ab)c = a(bc)$. One can then prove by induction that for all possible ways of grouping together the elements in the string $a_1 a_2 \cdots a_k$ with all resulting products ultimately being binary, the final answers come out the same. In such a grouping process, one inserts suitable parentheses into the spaces between the factors a_i . However, not every way of inserting parentheses is legitimate. For instance, ‘ $(a_1((a_2 a_3) a_4)$ ’ doesn’t make sense, while ‘ $((a_1 a_2) a_3) a_4$ ’ does. The difference between valid and invalid configurations of parentheses is that the valid ones satisfy the restriction mentioned in the problem, which simply ensures that (i) no grouping can close before it has opened, and (ii) no two groupings will overlap each other without one being contained within the other. In the present context, we are not interested in the factors a_i at all, but only in legitimate arrangements of parentheses. So, for instance, we think of ‘ $((a_1 a_2) a_3) a_4$ ’ as ‘ $((()))$ ’, and ‘ $((((a_1) a_2) (a_3 (a_4))))$ ’ as ‘ $((()))((()))$ ’.

Instead of demonstrating that this problem and the two previous problems satisfy the same recursion and initial condition, we show that they are equivalent by proving the existence of a bijection between arrangements of parentheses and valid handshake configurations, as introduced in Problem 2. Let the $2n$ people

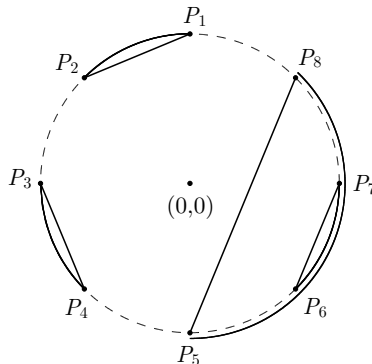
around the circular table be denoted by $P_1, P_2, \dots, P_{2n}$, ordered consecutively in the counterclockwise sense. Notice that for $k = 1$, our restriction says that there must be at least one opening parenthesis before the first closing parenthesis. Thus it makes sense to identify person P_1 with the first symbol in the parenthetical string, namely a ‘(’. The essential idea now is to identify each pair of handshake partners with a pair of parentheses that are ‘mates’. An opening parenthesis and a closing parenthesis are said to be *mates* in a given legitimate arrangement if the opening one appears to the left of the closing one, and between them there appear exactly as many opening parentheses as closing ones. So our bijection identifies handshake partners with parenthetical mates.

In order to prove that this correspondence is well defined and a bijection with a minimum of formality, we will transform everything into geometrical language. Represent the people $P_1, P_2, \dots, P_{2n}$ as the vertices of a regular $2n$ -gon inscribed in the unit circle $\mathcal{C}$. A handshake between P_i and P_j is then a chord P_iP_j of $\mathcal{C}$. A handshake configuration involves n non-crossing chords of $\mathcal{C}$. Now let us take all points (x, y) inside or on $\mathcal{C}$, except $(0, 0)$, and project them outward onto $\mathcal{C}$ along rays from the origin. The projection is given formally by

$$(x, y) \mapsto \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}} \right).$$

This maps each chord of $\mathcal{C}$ onto a minor arc or at worst a semicircle. Moreover, as illustrated in figure 2, the following statements are equivalent: (i) no two chords cross; (ii) no two corresponding arcs overlap without one being nested inside the other.

Figure 2: Correspondence between chords and arcs of $\mathcal{C}$.



We translate vertices P_i into parentheses as follows. Replace P_1 with ‘(’. Move counterclockwise around $\mathcal{C}$. If the current vertex is the first endpoint of its arc encountered so far, then replace it with ‘(’. Otherwise replace it with

‘)’). Read the resulting string of parentheses so that left-to-right corresponds to counterclockwise, starting from P_1 . Such a resulting string is clearly legitimate if it came from a non-crossing chord configuration. Conversely, if the resulting parenthetical string is legitimate, the chord configuration we started with must have been non-crossing.

For the chord configuration shown in figure 2, we get the following legitimate string of parentheses:

$$\begin{array}{cccccccc} P_1 & P_2 & P_3 & P_4 & P_5 & P_6 & P_7 & P_8 \\ (&) & (&) & (& (&) &) \end{array}$$

The bijection just described shows that the number of legitimate arrangements of parentheses is C_n , the n th Catalan number.

Problem 4. Let a ‘northeasterly lattice path’ be a sequence of lattice points in which any transition is either $(a, b) \rightarrow (a + 1, b)$ or $(a, b) \rightarrow (a, b + 1)$. Find the number of northeasterly lattice paths from $(0, 0)$ to (n, n) that do not go above the line $y = x$.

For this problem we give a bijection between admissible paths and legitimate arrangements of parentheses, as described in problem 3. Associate each opening parenthesis with a rightward move $(a, b) \rightarrow (a + 1, b)$, and each closing parenthesis with an upward move $(a, b) \rightarrow (a, b + 1)$. A northeasterly lattice path is admissible if and only if at no stage within it have there been more upward moves than rightward moves. This matches precisely the restriction on legitimate arrangements of parentheses stated in Problem 3. It follows that the number of admissible paths is also C_n .

Remarkably, as diverse as these four problems may have seemed *prima facie*, they have turned out to be governed by a single combinatorial phenomenon, namely the Catalan recursion.

3. EXPLICIT EXPRESSION FOR C_n . Consider the generating function

$$A = A(x) = \sum_{n=0}^{\infty} C_n x^n,$$

whose coefficients are the Catalan numbers. Then

$$\begin{aligned} A^2 &= \left(\sum_{i=0}^{\infty} C_i x^i \right) \left(\sum_{j=0}^{\infty} C_j x^j \right) \\ &= \sum_{n=0}^{\infty} \left[\sum_{i+j=n} C_i C_j \right] x^n = \sum_{n=0}^{\infty} \left[\sum_{i=0}^n C_i C_{n-i} \right] x^n. \end{aligned}$$

Using our recursive formula (1), and shifting the index of summation, we

obtain the equation

$$A^2 = \sum_{n=0}^{\infty} C_{n+1}x^n = \sum_{n=1}^{\infty} C_nx^{n-1},$$

and therefore

$$xA^2 = \sum_{n=1}^{\infty} C_nx^n.$$

This implies that $A = xA^2 + C_0 = xA^2 + 1$. Solving for A , we find that

$$A = \frac{1 \pm \sqrt{1-4x}}{2x}.$$

When $x = 0$, all terms of $A(x)$ except the constant term will zero out, so that $A(0) = C_0 = 1$. Since $A(x)$ is continuous within its interval of convergence, the limit of $A(x)$ as $x \rightarrow 0$ must be 1. It follows that the correct sign in the above formula is minus. That is,

$$A(x) = \frac{1 - \sqrt{1-4x}}{2x}. \quad (2)$$

We can expand the radical as a power series using *Newton's Generalization of the Binomial Theorem*, which states that

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k \quad (|x| < 1),$$

where α is any real constant, and $\binom{\alpha}{k}$ is given by the extended formula

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-k+1)}{k!}.$$

In our case, putting $\alpha = \frac{1}{2}$, we have

$$\begin{aligned} (1-4x)^{1/2} &= 1 + \sum_{k=1}^{\infty} \frac{(\frac{1}{2})(\frac{1}{2}-1)(\frac{1}{2}-2)\cdots(\frac{1}{2}-k+1)}{k!} (-1)^k (2)^{2k} x^k \\ &= 1 - \sum_{k=1}^{\infty} \frac{(1)(3)(5)\cdots(2k-3)}{2^k k!} 2^{2k} x^k \\ &= 1 - \sum_{k=1}^{\infty} \frac{(2k-2)!}{k! \{2^{k-1}(k-1)!\}} 2^k x^k \\ &= 1 - 2 \sum_{k=1}^{\infty} \frac{(2k-2)!}{k!(k-1)!} x^k \\ &= 1 - 2 \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^k. \end{aligned}$$

At the third step here, we restored the missing even factors $2, 4, \dots, 2k - 2$ to the numerator and denominator, so that the numerator was transformed into a factorial, $(2k - 2)!$. Substituting the above series expansion for $\sqrt{1 - 4x}$ in formula (2), we obtain

$$\begin{aligned}
A &= \frac{1 - \sqrt{1 - 4x}}{2x} \\
&= \frac{1}{2x} \left[1 - \left[1 - 2 \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^k \right] \right] \\
&= \frac{1}{x} \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^k \\
&= \sum_{k=1}^{\infty} \frac{1}{k} \binom{2k-2}{k-1} x^{k-1},
\end{aligned}$$

and so the coefficient of x^k is

$$\frac{1}{k+1} \binom{2(k+1)-2}{(k+1)-1} = \frac{1}{k+1} \binom{2k}{k}.$$

Since the function $A(x)$ was constructed so that the coefficient of x^n is the n th Catalan number C_n , we have finally proven that

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

4. COMBINATORIAL PROOF OF EXPLICIT FORMULA. The generating function method is straightforward and quite general, albeit tedious. However, we can also give a purely combinatorial proof of the explicit formula. To do this, we reinterpret Problem 3 in the following way: *How many zigzag paths are there from $(0, 0)$ to $(2n, 0)$ that never go below the x -axis?* Here, a *zigzag path* is a lattice path whose moves are either northeast or southeast, that is, either $(a, b) \rightarrow (a + 1, b + 1)$ or $(a, b) \rightarrow (a + 1, b - 1)$. Such paths with the added restriction of never going below the x -axis are clearly equivalent to legitimate strings of parentheses, where each northeast move is considered an opening parenthesis, and each southeast move a closing parenthesis.

The number of *unrestricted* zigzag paths from $(0, 0)$ to $(2n, 0)$ is just $\binom{2n}{n}$, since we have $2n$ horizontal units to travel, and all we have to decide is which n of these $2n$ rightward steps will result from a northeast move. There must be n northeast moves and n southeast moves, since our path starts and ends at the same height $y = 0$. We will now try to compute the number of zigzag paths from $(0, 0)$ to $(2n, 0)$ that go below the x -axis at least once.

If a path goes below the x -axis, it must go through at least one point with a y -coordinate of -1 . Consider the first (leftmost) point of such a path in which

the y -coordinate is -1 , say $(a, -1)$. Reflect the portion of the path between $(0, 0)$ and $(a, -1)$ through the line $y = -1$, leaving the rest of the path fixed. This produces a path from $(0, -2)$ to $(2n, 0)$. Since $(a, -1)$ is uniquely defined, there is a bijection between zigzag paths from $(0, 0)$ to $(2n, 0)$ that go below the x -axis and *unrestricted* zigzag paths from $(-2, 0)$ to $(2n, 0)$.

The number of paths of the latter kind is $\binom{2n}{n-1}$, since the $2n$ rightward steps must be divided into $n+1$ northeast moves and $n-1$ southeast moves. [If N and S are numbers of northeast and southeast moves, then $N+S=2n$ and $N-S=0-(-2)=2$.] Thus

$$\begin{aligned} C_n &= \binom{2n}{n} - \binom{2n}{n-1} \\ &= \frac{(2n)!}{n!n!} - \frac{(2n)!}{(n+1)!(n-1)!} \\ &= \frac{(2n)!}{(n-1)!(n-1)!} \left(\frac{1}{n^2} - \frac{1}{n^2+n} \right) \\ &= \frac{(2n)!}{(n-1)!(n-1)!} \cdot \frac{1}{n^2(n+1)} \\ &= \frac{1}{n+1} \cdot \frac{(2n)!}{n!n!} \\ &= \frac{1}{n+1} \binom{2n}{n}. \end{aligned}$$

5. CONCLUSION. Catalan numbers are a prominent feature of the landscape in combinatorial mathematics. The problems discussed above represent only a few scenarios in which they arise. For many further details and applications, the interested reader is directed to [1, 2, 3, 4].

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The Nonagon Anomaly

Katherine Banks[†]

1. INTRODUCTION. In this paper, we prove using combinatorial methods that a 9-sided convex lattice polygon cannot have exactly 8 or 9 interior lattice points. This result, known as the *Nonagon Anomaly*, has been proven before in a heavily geometric manner that works only for the 2-dimensional case. Our proof shows the Nonagon Anomaly to be true for any set of points S satisfying the *Pentagon Property*, namely that a convex pentagon whose vertices lie in S must have at least one point of S in its interior. The integer lattice in $\mathbb{R}^2$ is known to satisfy the Pentagon Property, so we still phrase all arguments in terms of convex lattice polygons for notational ease and familiarity. Other sets satisfying the Pentagon Property, as we show, include planar polygons in the n -dimensional integer lattice.

We first re-prove the *Fat Hexagon Theorem*, known for the 2-dimensional integer lattice, which states that a convex lattice hexagon with a lattice point that is not a vertex on its boundary must have at least two interior lattice points. The new proof holds for any set of points satisfying the Pentagon Property. We proceed to prove that a convex lattice pentagon with at least three non-vertex lattice points on its boundary must have two or more interior lattice points, and introduce other lemmas before moving to casework. Finally, we show briefly that there are no anomalies for $n \leq 7$.

2. BACKGROUND. A *lattice point* is a point with integral coordinates. A *lattice polygon* is a polygon whose vertices are lattice points. The study of lattice polygons has been undertaken by several authors from widely divergent points of view, ranging from elementary geometry [2, 4, 7, 8, 9, 11, 12] to algebraic geometry [6, 13]. The attraction to the field for many researchers seems to stem from an affinity for doodling polygons that can become something of an obsession; the reader should consider himself fairly warned. Lattice polygons are easily visualized, and through activities like the Geoboard game Coleman discusses in [2] can be investigated with very little mathematical background required. Recently, lattice geometry has advanced with the help of computer searches, heavily used by Rabinowitz, which can allow researchers to formulate conjectures that are later proven mathematically.

The study of lattice polygons has been popularized in the form of Pick's theorem relating the area, $A(K)$, of a simple lattice polygon K in 2 dimensions to

[†]This is a modification of Ms Banks' entry in the 2007 Intel Science Talent Search, which was awarded fourth place overall. — Ed.

the number of vertices, the number of lattice points on its boundary (excluding the vertices), and the number in its interior, usually denoted $v(K)$, $b(K)$ and $g(K)$ respectively. Pick showed that

$$A(K) = g(K) + \frac{1}{2}\{v(K) + b(K)\} - 1.$$

Since then, numerous proofs have been given, and the formula has been adapted for non-convex and non-simple polygons. Most work has been restricted to polygons in the 2-dimensional integer lattice, since there is no generalization of Pick's theorem giving the volume of a lattice polyhedron as a function of the number of boundary and interior lattice points (this fact is noted in [11], since the tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, k)$ has volume $k/6$ dependent on k and no boundary or interior lattice points). Scott proved the inequality $v(K) + b(K) \leq 2g(K) + 7$ using elementary techniques [12], and the alternate proofs of this result given by other researchers [13] have opened up the study of lattice polygons to methods of algebraic geometry.

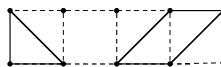
Much work has been done in finding equations and inequalities relating $A(K)$, $b(K)$, $g(K)$, and $v(K)$, involving both metric and incidence properties of lattice polygons. The notion of *lattice equivalent* polygons has been formalized, a necessity given that rigid motions of the plane do not preserve the lattice point content of polygons, and some attention has been paid to polygons with vertices in the triangular and hexagonal lattices [3].

3. ACHIEVABLE NUMBERS OF INTERIOR LATTICE POINTS.

The earliest work toward determining the values of the function $g(n)$, the minimum number of lattice points that must be in a convex lattice n -gon in the plane, was done by Arkinstall in [1]. Note that $g(n)$ is distinct from the function $g(K)$, which has as its domain the set of convex lattice polygons and gives the number of lattice points in the interior of a specific polygon. From now on, we will use “polygon” to mean “convex lattice polygon,” “pentagon” for “convex lattice pentagon,” etc., unless otherwise noted.

Arkinstall gave geometric proofs that a pentagon must have at least one interior lattice point, and that a hexagon with a lattice point that is not a vertex on its boundary must have at least two interior lattice points. In other words, $g(5) = g(6) = 1$. We can see in Figure 1 that $g(3) = g(4) = 0$, but Arkinstall showed that all trapezia (quadrilaterals with no pair of sides parallel) must have an interior lattice point. We will not take up such matters here.

Figure 1: $g(3) = g(4) = 0$.



Beginning with Arkinstall's theorem, Rabinowitz has used computer search to create a census of all polygons with one interior lattice point up to "lattice equivalence," a concept which he defines that essentially considers two polygons with the same lattice point content on their boundaries and interiors equivalent. From there, noting that $g(n)$ is clearly monotonic and increasing, he and others have used a combination of computer search and geometric and combinatorial methods to find the value of $g(n)$ for a few values of n , and bounds on $g(n)$ in some other cases.

It is tempting to think that there must be a convex lattice n -gon containing g interior lattice points for all $g \geq g(n)$, but in fact computer search has shown this to be untrue. In particular, Rabinowitz gives a proof in [9] that no convex lattice octagon can have exactly five interior lattice points, and Kolodziejczyk in [5] gives a heavily geometric proof of Rabinowitz's conjecture that no 9-sided polygon in the 2-dimensional integer lattice can contain exactly eight or nine points in its interior, the so-called Nonagon Anomaly, and similarly that no decagon can have eleven or twelve interior lattice points.

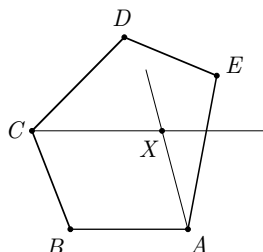
In addition to his geometric work on lattice polygons, Rabinowitz has proven some related results in a slightly more general setting by combinatorial methods. In particular, he proved the Octagon Anomaly for any set of points S satisfying the Pentagon Property: any convex pentagon with vertices in S must have a point of S in its interior. As Arkinstall's theorem shows, the set of polygons with vertices in the 2-dimensional integer lattice satisfies the Pentagon Property, so for notational convenience and familiarity we will continue to speak of lattice polygons, but the proofs we give of results about these polygons in the rest of this paper hold without modification for any set of points satisfying the Pentagon Property. Our chief aim will be to present a simpler, combinatorial proof of Kolodziejczyk's Nonagon Anomaly that holds for polygons in this somewhat more general setting, and to discuss other anomalies (unachievable numbers of interior lattice points above the minimum) for these polygons.

The Nonagon Anomaly. *A convex nonagon with vertices in a point set S satisfying the Pentagon Property cannot have exactly 8 or 9 elements of S in its interior.*

4. POINT SETS SATISFYING THE PENTAGON PROPERTY. We reproduce here Arkinstall's proof that the 2-dimensional integer lattice satisfies the Pentagon Property to acquaint the reader with common lines of reasoning in lattice geometry. Note that this proof holds without modification for planar polygons in the n -dimensional integer lattice, so we have already found sets of points satisfying the Pentagon Property for which Pick's Theorem and other standard geometric tools used on the 2-dimensional lattice do not hold.

Arkinstall's Lattice Pentagon Theorem (Arkinstall). *A convex lattice pentagon must have at least one interior lattice point.*

Figure 2:



Proof. Let $ABCDE$ be a planar, convex lattice pentagon in n -space. Since the sum of the interior angles of a pentagon is 3π , the sum of the 5 pairs of adjacent interior angles is 6π , so some pair of adjacent angles, say A and B (Figure 2) must sum to more than π . Assume that point C is not further from line AB than point E , and take AB to be horizontal. Draw the parallel to BC through A , and the parallel to AB through C ; call their intersection X . Ray AX lies inside angle A , since $m\angle A + m\angle B > \pi$ and $ABCX$ is a parallelogram. Similarly, ray CX lies inside angle C , so X lies in the intersection of the interiors of angles A and C . Since the vertical distance from C to AB is equal to the vertical distance from X to AB and less than that from E to AB , X lies in the interior of $ABCDE$. But if three vertices of a parallelogram are lattice points, the fourth vertex must also be a lattice point. (Note that this holds for n -space, since the coordinates of X are found by adding vectors AB and BC , both of which have integer components). Thus X is a lattice point that will always fall inside pentagon $ABCDE$. $\square$

We will henceforth abbreviate the Lattice Pentagon Theorem as the ‘LPT’.

Rabinowitz [10] conjectures that all point sets in $\mathbb{R}^n$ satisfying the Pentagon Property can be transformed into $\mathbb{Z}^2$ by an *integral unimodular transformation* $\mathbf{x}' = A\mathbf{x}$, where A is an $n \times n$ matrix with integral entries and $\det(A) = 1$, though he gives no examples of such sets.

Certainly some of the geometric arguments we make for the 2-dimensional lattice can generalize to n -space, but because Pick’s theorem fails even for planar subsets of the n -dimensional lattice, and because we lack a complete characterization of what sets satisfy the Pentagon Property, we cannot use Pick’s Theorem in our proof and must dispense with many of the geometric methods Rabinowitz and Arkinstall use. To see that we cannot give a formula relating area to the number of interior and boundary lattice points of a polygon in a planar subset of $\mathbb{Z}^n$, note that the triangle in $\mathbb{Z}^3$ with vertices $(0, 0, 0)$, $(1, 0, 0)$ and $(1, k, 1)$ is a primitive triangle (that is, has no interior or boundary lattice points besides

its vertices) with area $\frac{1}{2}\sqrt{k^2 + 1}$ dependent on k . Pick's theorem relies on the fact that every primitive triangle in the 2-dimensional lattice has area $\frac{1}{2}$.

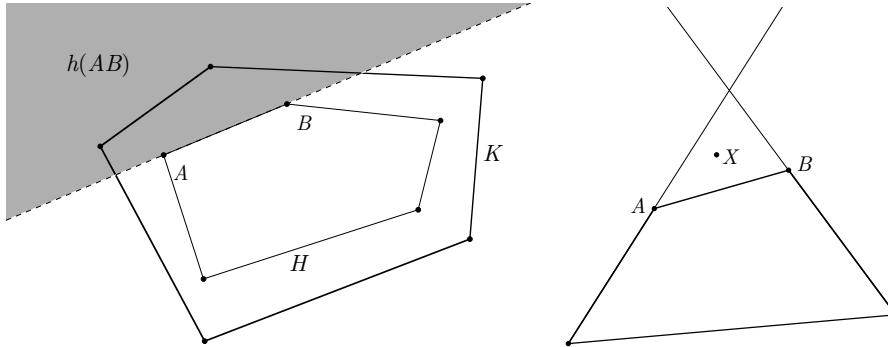
5. PRELIMINARY THEOREMS AND NOTATION. Let $g(K)$ denote the number of lattice points in the interior of a polygon K , and $b(K)$ the number of lattice points on the boundary of K , excluding the vertices. The following are theorems previously proven by Arkinstall [1]. and Rabinowitz [7, 8] which are instrumental in our proof of the Nonagon Anomaly for all point sets satisfying the Pentagon Property. Though they are phrased in terms of lattice points, all theorems stated here hold for any point set satisfying the Pentagon Property. We also introduce here the notation we will be using throughout our proof and some new definitions.

Definition: Interior Hull. *The interior hull of a convex lattice polygon K is the convex hull of the lattice points in the interior of K .*

Definition: Lean. *K is lean if there are no lattice points on its boundary that are not vertices. If it is not lean, we will call it fat.*

Notation: $h(AB)$. *Let H be the interior hull of K , with an edge AB . Then $h(AB)$ denotes the open halfplane of line AB containing no lattice points which are in the interior of K . We may also refer to $h(AB)$ as the open exterior halfplane of AB . See the leftmost figure below.*

Figure 3: Left, $h(AB)$. Right, “to itself.”



2-Vertex Restriction. *Let K be a convex lattice polygon with interior hull H . Let AB be an edge of H . Then $h(AB)$ contains at most two vertices of K .*

Proof. Suppose $h(AB)$ contained three vertices of K , X , Y and Z . Consider points A , B , X , Y and Z . No one of the vertices of K can be contained in the convex hull of the other four points, since it would then be an interior point of K and not a vertex. Thus pentagon $BAXYZ$ is convex and satisfies the Pentagon Property. Then there must be an interior lattice point in $BAXYZ$ that lies entirely in $h(AB)$, contradicting the definition of $h(AB)$. $\square$

The Interior Hull Vertex Inequality. Let K be a convex lattice polygon with $H = H(K)$ the interior hull of K . If $v(K) \geq 7$, then $v(K) \leq 2v(H)$.

The Interior Hull Boundary Inequality. Given polygon K with interior hull H , if $v(K) \geq 9$, then $2v(K) \leq 3(v(H) + b(H))$.

Definition: To Itself. Given a polygon K and its interior hull H , with a side AB and adjacent sides AD and BC . We will say AB , or $h(AB)$, has a vertex of K to itself if that vertex is contained in $h(AB)$ but not $h(AD)$ or $h(BC)$. For example, in Figure 3, above right, $h(AB)$ has point X to itself.

Definition: Crowded. A side AB of the interior hull of K is crowded if $h(AB)$ contains two vertices of K to itself.

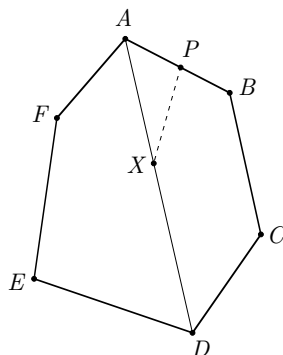
Definition: Shared. Consider two adjacent sides, AB and BC , of the interior hull H of polygon K . The region that is the intersection of $h(AB)$ and $h(BC)$ will be called the shared region between these two sides.

Note that every vertex of K must either be such that some side of H has that vertex to itself, or that vertex is contained in one of the shared regions exterior to H . Each side of H has associated with it one region that it has to itself and two regions adjacent to that region which are shared. Since no open exterior halfplane of H can have more than two vertices of K , a crowded side cannot have any vertices of K in one of its shared regions.

6. LEMMAS. Here we give new lemmas that will aid our proof and re-prove Rabinowitz's Fat Hexagon Theorem combinatorially for our use. Note that this is a new proof, not the one originally given by Rabinowitz [7].

Fat Hexagon Theorem. A fat convex lattice hexagon contains at least two interior lattice points.

Figure 4:

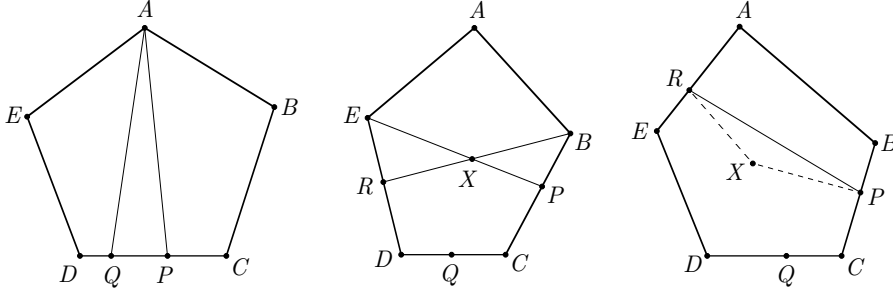


Proof. Consider a convex lattice hexagon $K = ABCDEF$, as shown in Figure 4, with the lattice point P on AB a non-vertex lattice point on the hexagon's boundary. Suppose for the sake of contradiction that K has exactly one interior lattice point, X . Then X must lie on diagonal AD , for if it were to the left $XABCD$ would be an empty, convex lattice pentagon in violation of the LPT, and similarly $XAFED$ would violate the LPT if X were to the right of AD . But then the convex lattice pentagon $PXDCB$ would violate the Lattice Pentagon Theorem. Thus, a fat convex lattice hexagon must contain at least two interior lattice points. $\square$

Fat Pentagon Lemma. *If pentagon K has $b(K) > 2$, then $g(K) \geq 2$.*

(Equivalently, a convex lattice pentagon with one interior lattice point can have at most two non-vertex lattice points on its boundary. This second formulation will be most useful to us later.)

Figure 5: A pentagon with $b > 2$ must have $g > 1$.



Proof. See Figure 5. Consider a convex lattice pentagon $K = ABCDE$ with a single interior lattice point X . First, suppose that there were two non-vertex boundary lattice points, P and Q , on side CD of K . Now X cannot lie to the left of diagonal AQ , since then $AXQCB$ would contradict the LPT, and similarly it cannot lie to the right of AQ . Thus, X lies on AQ . By the same reasoning, X must lie on AP . Since $X \neq A$, this is impossible, and a pentagon with $g(K) = 1$ cannot have two non-vertex boundary lattice points on one side.

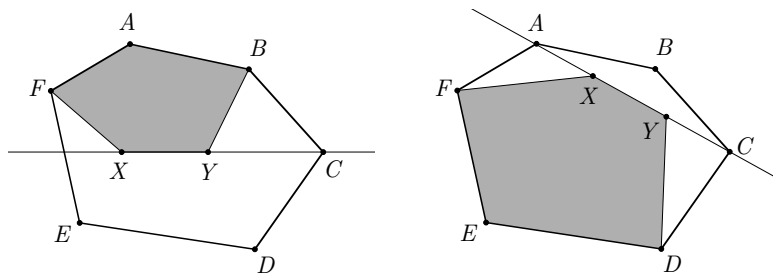
Now suppose that K had one non-vertex boundary lattice point on each of three adjacent sides, say P on BC , Q on CD , and R on DE (see the second figure). Once again, we find that the single interior lattice point X must lie on both diagonals EP and BR , and thus at their intersection. But then pentagon $RXPCD$ contradicts the Pentagon Property.

Finally, suppose the three non-vertex boundary lattice points were such that P is on BC , Q on CD , and R on EA . Then X must lie in the interior of pentagon $ERPCD$. But then $ABPXR$ forms a convex lattice pentagon, contradicting

the Pentagon Property. (Note that since we have not made use of the point Q , this also proves that there does not exist a pentagon with two non-vertex lattice points on its boundary, one on each of two non-adjacent sides). This covers all cases, and thus a pentagon with three non-vertex lattice points on its boundary must have more than one lattice point in its interior. $\square$

Lemma 2. *Let K be a convex lattice polygon with $v(K) = 6$ and all interior lattice points collinear. Then the line containing the interior lattice points passes through two opposite vertices of K . Moreover, the two vertices are opposite ones.*

Figure 6: When all interior lattice points are collinear for a hexagon, the line containing them passes through two opposite vertices.



Proof. Suppose the line containing the interior lattice points passed through less than two vertices of K . Since $v(K) = 6$, one of the halfplanes of the line would then contain at least three vertices of K . These vertices, together with any two of the interior lattice points, would form a lattice pentagon with no interior lattice points, contradicting the LPT. See above left, Figure 6.

Now suppose $K = ABCDEF$ and the line containing the interior lattice points passed through two vertices that were not opposite (two vertices apart), say A and C . Then $ACDEF$ would be an empty lattice pentagon, a contradiction. So the line containing the interior lattice points must pass through two opposite vertices of H , say A and D . $\square$

7. REDUCING THE NUMBER OF CASES. Let K be a convex lattice nonagon, $v(K) = 9$, with interior hull H . As in Kolodziejczyk's strategy, we must consider all possible arrangements of either 8 or 9 lattice points in the interior of K and show that each leads to a contradiction. Note that all lattice points interior to K must be either vertices of H , non-vertex lattice points on the boundary of H , or lattice points in the interior of H (by definition of H , there are no lattice points in the interior of the region bounded by H and K).

By the Interior Hull Vertex Inequality, we know that $v(K) \leq 2v(H)$, so $v(H) \geq 5$ for a 9-gon. Rabinowitz showed that a heptagon must have at least 4 interior lattice points. Since we are considering only 9-gons with 8 or 9 interior lattice points and attempting to show they do not exist, we need only look at the cases when $v(H) = 5$ or 6, since $v(H) \geq 7$ would imply $g(K) \geq 11$, and the function $g(n)$ is nondecreasing everywhere. Thus we will consider only the cases when H is a pentagon or hexagon. Furthermore, by the Interior Hull Boundary Inequality $v(H) + b(H) \geq 6$, so if $v(H) = 5$, $b(H) \geq 1$, and thus if H is a pentagon it is fat.

8. WHEN H IS A PENTAGON. Since the points we are considering satisfy the Pentagon Property, $g(H) \geq 1$. By the Fat Pentagon Lemma, there exists no convex lattice pentagon with $b(H) = 3$ and $g(H) = 1$, so we need not consider that configuration of interior points. All potential configurations of the interior points of K remaining for the case when the interior hull is a pentagon have $b(H) < 3$. In these other cases, pentagons with, for example, $b = 1$ and $g = 3$ do exist, so in reaching a contradiction we must rely on the fact that the pentagons under consideration are the interior hulls of a nine-sided convex lattice polygon.

Lemma 3. *If K is a nonagon and its interior hull H is a pentagon, then at least three sides of H must be crowded.*

Proof. To show that the pentagon H must have at least three crowded sides, let us determine what we can about the distribution of the nine vertices of K around the exterior of H .

By the Two-Vertex Restriction, we know that the sum of the number of vertices contained in the open exterior halfplanes of each side is at most 10. This number may include some vertices that are “double-counted” because they are contained in the shared region of some pair of sides. We also know that the total number of vertices of K in the exterior of H is 9.

Let s be the number of vertices that are contained in exactly one open exterior halfplane; these are the vertices that are single-counted when we sum the number of vertices in each of the five exterior halfplanes. Let d be the number of vertices contained in some shared region exterior to H ; these vertices are double-counted. From what we noted above, we have $s + 2d \leq 10$ and $s + d = 9$. Thus, we can conclude that $d = 0$ or $d = 1$. If $d = 0$, then all nine vertices of K are single-counted, and thus each one is contained in the region that some side of H has to itself. Then four sides of H would be crowded, since all 9 vertices of K are either contained in a shared region or a region had to itself by one side of H .

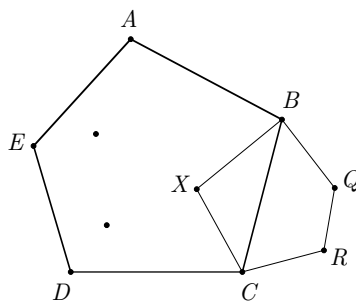
If $d = 1$, then there are still eight vertices of K that are contained in exactly one open exterior halfplane of some side of H . Now H has five sides, each with at most two vertices to themselves, and the two sides whose exterior halfplanes form the shared region with a vertex of K in it have at most one vertex of K to themselves. If neither or just one had a vertex of K to itself, there would be at least seven vertices to place among the remaining three exterior halfplanes,

an impossibility because of the Two-Vertex Restriction. If both halfplanes had a vertex of K to themselves, there would be six vertices remaining to distribute among three sides, none of which has a vertex in either of its shared regions, and so all three sides must be crowded.

Thus, we have shown that any pentagon that is the interior hull of a nonagon must have at least three crowded sides. $\square$

Let $H = ABCDE$ be the interior hull of nonagon K with $b(H) < 3$. Because $v(K) = 9$, by Lemma 3 at least three sides of H are crowded. Thus some side of H , say BC , is crowded and lean. Let Q and R be the two vertices of K that BC has to itself. See Figure 7.

Figure 7:



Let the closest interior lattice point to line BC be X . If there is more than one such point equidistant from BC , simply let X be any one of them. Note that the quadrilateral $BCRQ$ must be convex, since if, say, Q were contained in $\triangle BCR$, then K would not be convex. Thus $XBQRC$ forms a lattice pentagon with no interior lattice points, contradicting the Pentagon Property.

This shows that the interior hull of a nonagon cannot be a pentagon with $b < 3$. Since all possible configurations of 8 or 9 lattice points inside a nonagon with a pentagonal interior hull would have $b(H) < 3$, this shows that no nonagon with a pentagonal interior hull can have exactly 8 or 9 interior lattice points.

9. WHEN H IS A HEXAGON. When $v(H) = 6$, it is possible that H is lean. The cases we must consider and dispose of are $b(H) = g(H) = 1$, $b(H) = 2$ and $g(H) = 1$, $b(H) = 0$ and $g(H) = 3$, $b(H) = 0$ and $g(H) = 2$, and $b(H) = 1$ and $g(H) = 2$. These are the only possible cases for $v(H) = 6$ and $g(K) = 8$ or 9 .

When $b(H) = 1$ or 2 , it is not possible to have $g(H) = 1$, since by the Fat Hexagon Theorem, all fat hexagons have at least two interior lattice points. We are thus left with three possible configurations of the interior lattice points to dispose of: $b(H) = 0$ and $g(H) = 2$ or 3 , and $b(H) = 1$, $g(H) = 2$. Since these

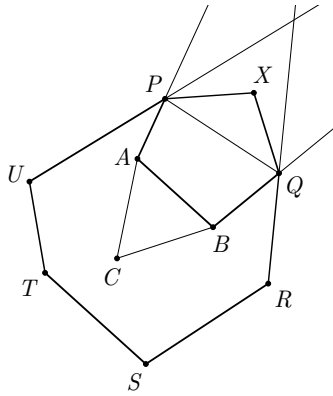
remaining configurations can exist as convex lattice hexagons (that is, there exist, for example, convex lattice hexagons with $b(H) = 0$ and $g(H) = 2$), we must rely on the fact that these particular hexagons are the interior hulls of a 9-sided convex lattice polygon.

Case when $b(H) = 0$ and $g(H) = 3$:

Let $H = PQRSTU$ be a lean hexagon, with lattice points A, B, C in its interior. Let us first consider the case when A, B, C are not collinear, Figure 8.

Since $v(H) = 6$ and $h(AB)$, $h(BC)$, and $h(AC)$ can each contain at most two vertices of H by the Two-Vertex Restriction, all three sides must be crowded. (If some were not, those vertices would double-count toward the Two-Vertex Restriction and not allow H to have six sides). The two vertices of H contained in a single exterior halfplane of a side of $\triangle ABC$ must be adjacent ones.

Figure 8:

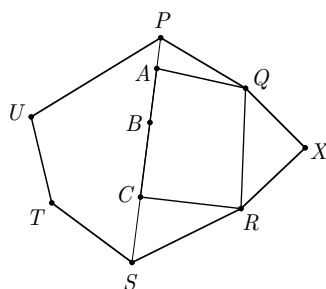


Consider AB , and let P and Q be the vertices of H contained in $h(AB)$. We have already shown that AB must contain P and Q to itself. Let X be a vertex of K that $h(PQ)$ has to itself (we can say this without loss of generality because at least one of the three sides of H contained in an exterior halfplane of $\triangle ABC$ must have a vertex of K to itself, for otherwise K could have at most six vertices). Now $ABQP$ is a convex quadrilateral. Since X is contained in the region $h(PQ)$ has to itself, it is contained in the region of $h(PQ)$ bounded by rays AP and BQ . Thus, $ABQXP$ is a convex pentagon, contradicting the Pentagon Property.

Next, consider the case when lattice points A, B and C inside hexagon H are collinear, Figure 9. By Lemma 2, line AB passes through two opposite vertices of H , say P and S . At least one of sides QR and TU , the sides not

containing any point of line PS on them, must have a vertex X of K to itself. Otherwise, K could have at most 8 vertices distributed among the four other exterior halfplanes, but it has nine. Say QR contains X to itself. Then, since H is lean, $AQXRC$ forms a convex, empty lattice pentagon in violation of the Pentagon Property.

Figure 9:



Case when $b(H) = 0$ and $g(H) = 2$:

In this case, the interior lattice points of H must be collinear. The reasoning from the second part of the previous case shows this case to be impossible.

Case when $b(H) = 1$ and $g(H) = 2$:

Let $H = PQRSTU$ be the interior hull of K . Again, we have that the lattice points A and B interior to H must be collinear, and the line containing them must pass through two opposite vertices, P and S , of H (by Lemma 2). Let X be the lattice point on the boundary of H that is not a vertex. See Figure 10.

If X is not on QR or TU , the two sides of H not containing points P or S of the opposite diagonal, then both of those sides are lean since X is the only non-vertex lattice point on the boundary of H . If neither contained a vertex of K to itself, by the Two-Vertex Restriction we could place at most 8 vertices of K in the exterior halfplanes corresponding to the remaining four sides. Thus, in this case, at least one of QR or TU is lean and has a vertex of K to itself. The same contradiction is reached as in the previous case.

So suppose X is on QR . Note that TU must not have any vertex of K to itself, for if it did, the same contradiction as above would be reached.

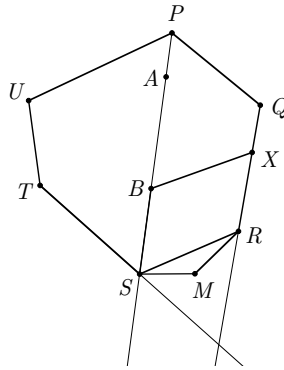
Again, let s be the number of vertices of K that are single-counted toward the Two-Vertex Restriction (i.e., are contained in exactly one open exterior half-plane of H), and d be the number of vertices of K contained in some region ex-

terior to H that is the intersection of two open exterior halfplanes. By the Two-Vertex Restriction, we then have $s + 2d \leq 12$, and we know $s + d = v(K) = 9$, so $d \leq 3$, $s \geq 6$, and thus some side of H must be crowded.

If the side of H that is crowded is also lean (that is, it is any side but QR), the two vertices of K contained in that side, the endpoints of that side, and the closest interior lattice point to that side form a convex pentagon with no interior lattice points, contradicting the Pentagon Property.

So let QR be crowded, thus with two vertices of K to itself. By supposition TU has no vertex of K to itself. If another side of H had no vertex of K to itself, by the Two-Vertex Restriction we could place at most six of the seven unplaced vertices of K in the remaining three open exterior halfplanes. Thus every side of H has at least one vertex of K to itself except TU , which by hypothesis has none, and QR has two vertices to itself. Let M be the vertex of K that side RS has to itself (see figure). Pentagon $XRMSB$ is convex, since the region RS has to itself is contained in the region external to H and bounded by line RS and rays QR and PS . But $XRMSB$ has no interior lattice points, violating the Pentagon Property.

Figure 10:



We have now shown that a nonagon with vertices in a point set satisfying the Pentagon Property cannot have exactly 8 or 9 points of that set in its interior.

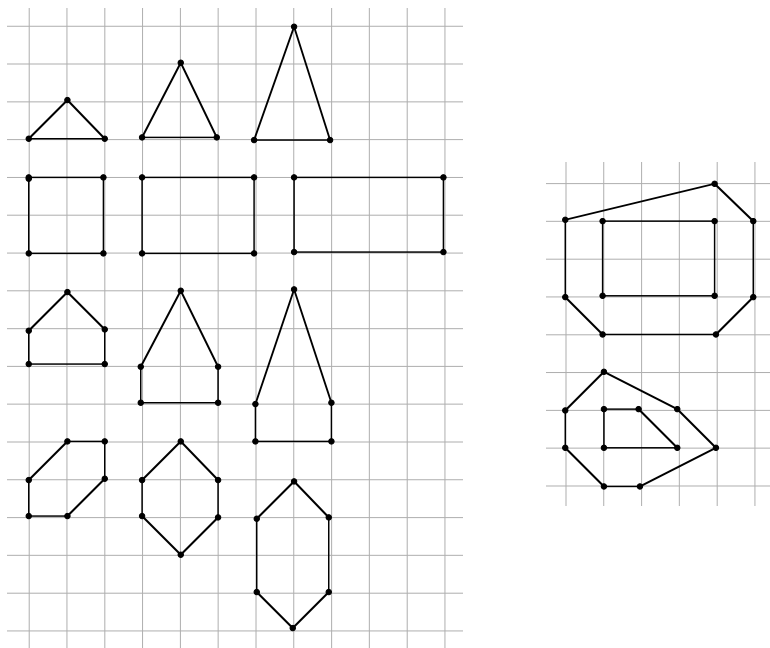
10. OTHER ANOMALIES. We are confident that the Decagon Anomaly [5] can be proven more generally for any point-set satisfying the Pentagon Property with combinatorial methods and casework similar to what we have done here.

1. For $n < 8$, there can be no anomalies for n -gons that result only from consequences of the Pentagon Property, because for convex lattice polygons in the 2-dimensional lattice with these numbers of sides every number of interior points above the minimum is achievable.

2. For $n < 7$, all values are achievable because the interior lattice points can be collinear and a simple algorithm takes a polygon gives a new polygon with one additional lattice point in its interior that is collinear with the rest (Figure 11).
3. In particular, starting from the triangle with vertices $A(0,0)$, $B(1,1)$ and $C(2,0)$, we can generate $\triangle AB'C$ with the y -coordinate of B' increased by one at each iteration; this procedure adds one interior lattice point each time, and generates every number of interior lattice points above zero.
4. For $n = 4$, for a given number of interior lattice points $I = ab$, a rectangle of length $a + 1$ and width $b + 1$ will have I interior lattice points. The rectangle $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$ has no interior lattice points, giving a 4-gon with the minimum number of interior lattice points. Alternately, construct the rectangle $(0,0)$, $(k+1,0)$, $(k+1,1)$, $(0,1)$, which can contain any number of interior lattice points above 1 as k ranges over the nonnegative integers.
5. For $n = 5$, construct the pentagon $A(0,0)$, $B(0,1)$, $C(1,2+k)$, $D(2,1)$, $E(2,0)$, and this pentagon will take on any number of interior lattice points above the minimum of 1 as k ranges over the nonnegative integers.
6. Similarly, for $n = 6$, construct the same pentagon as in the $n = 5$ case, adding a vertex $F(1, -m - 1)$ and letting m range over the nonnegative integers as well. The centrally symmetric hexagon $A(0,1)$, $B(1,1)$, $C(1,0)$, $D(0,-1)$, $E(-1,-1)$, $F(-1,0)$ has the minimum single interior lattice point, as noted by Rabinowitz.
7. For $n = 7$ we must take a little more care, since the interior lattice points cannot be collinear. Let I be the desired number of interior lattice points; we know $I \geq 4$. See Figure 11.
 - (a) If I is composite, we can form a rectangle that has I lattice points between its interior and its boundary, and then construct a convex heptagon with that rectangle as its interior hull. In particular, rectangle $A(0,0)$, $B(a,0)$, $C(a,b)$, $D(0,b)$ will have ab interior lattice points for $a, b > 1$. Then construct the heptagon with vertices $(-1,0)$, $(0,-1)$, $(a,-1)$, $(a+1,0)$, $(a+1,b)$, $(a,b+1)$, and $(-1,b)$; this heptagon is convex and has $ABCD$ as its interior hull.
 - (b) If $I = p$ is prime, we construct a trapezoid in a similar fashion, making sure that no additional lattice points are captured. In particular, let $p = 2a + 1$ for some a (the heptagon cannot have only two interior lattice points, so since p is prime and $p > 2$ it is of this form). Construct trapezoid $A(0,0)$, $B(a+1,0)$, $C(a,1)$, $D(0,1)$, which has no lattice points in its interior and p , including the vertices, on its boundary. Now construct the heptagon with vertices $(-1,0)$, $(-1,1)$, $(0,2)$, $(a+1,1)$, $(a+2,0)$, $(a,-1)$, and $(a-1,-1)$. This heptagon is convex and has exactly p interior lattice points.

Thus, there are no heptagon anomalies; there exists a convex lattice heptagon with g interior lattice points for all $g \geq g(7) = 4$.

Figure 11: Left, $n < 7$. Right, $n = 7$.



11. FURTHER RESEARCH. We hope that the techniques used to prove the Nonagon Anomaly here can later be developed into methods applicable to the general problem of mathematically proving the existence of anomalies. The approaches so far have basically been casework of varying levels of tedium, so more generally applicable methods would be welcome, with the ultimate goal being a complete characterization of the achievable number of interior lattice points for a given number of sides.

While we have studied sets of points satisfying the Pentagon Property, it may be fruitful to study polygons with vertices in other lattices, for example the equilateral triangular or hexagonal lattices. One could also place further geometric restrictions on the types of polygons studied and determine which numbers of interior lattice points are achievable for those specific types of polygons. A complete characterization of point sets satisfying the Pentagon Property might illuminate the connection between the combinatorial approach we have taken here and the geometric methods used previously.

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A Note on Union-Closed Families

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1. INTRODUCTION. A finite family of sets is said to be *union-closed* if the union of any two of its members also belongs to the family. Let $\mathcal{F}$ be a finite union-closed family of sets with $f := |\mathcal{F}|$. [If X is a finite set, then we write $|X|$ for the cardinality of X .] Peter Frankl conjectured in 1979 that for any such $\mathcal{F}$, there exists an element $a \in \bigcup_{S \in \mathcal{F}} S$ which appears as a member in at least $f/2$ sets of $\mathcal{F}$. This conjecture has come to be known as the *union-closed sets conjecture* or *UCS conjecture*, and at present remains unproven. In what follows, we write $L := L(\mathcal{F}) := \bigcup_{S \in \mathcal{F}} S$.

Poonen [1] has shown (cf. Theorem 1) that as far as the UCS conjecture goes, no generality is lost if we restrict our attention to cases in which L is finite. In such cases, $L \in \mathcal{F}$, since L can be built inductively by a process in which two-fold unions of sets in $\mathcal{F}$ are taken finitely many times (whereas if L is infinite, one may need an infinite union of sets in $\mathcal{F}$ to produce L). Whenever L is finite, we let $\ell := |L|$.

Since 1979, there has been much work on the subject of union-closed sets. The conjecture has gradually been proven true for all $\mathcal{F}$ with $f \leq 40$, as well as all $\mathcal{F}$ with $\ell \leq 11$. Poonen [1] and others have also shown that the conjecture holds for any $\mathcal{F}$ that contains a set of 1 or 2 elements.

Wójcik [2] defined a notion of ‘density’ for union-closed families and proved certain results regarding the minimum density of families whose sets together involve ℓ elements. We build on these results to prove that for any $\mathcal{F}$ with $\ell \geq 4$, some element $a \in L$ must appear in at least

$$\frac{2f}{1 + \sqrt{2\ell + 1}}$$

sets of $\mathcal{F}$.

2. DEFINITIONS AND NOTATIONS. We use the following alphabetic conventions. *Ur-elements*, basic objects which are not themselves sets, will be denoted by $a, b, c, \dots$. *Basic sets*, whose only members are ur-elements, will be denoted by $S, T, U, V, \dots$. *Families* or *collections* of basic sets will be denoted by $\mathcal{A}, \mathcal{B}, \dots, \mathcal{F}, \dots$. Finally, aggregates of families of basic sets will be denoted by $\Gamma, \Delta, \dots, \Lambda, \dots$.

[†]This paper is a modification of Mr Balla’s entry in the 2007 Intel Science Talent Search. The current version is the result of a collaboration with Mr Stern, faculty editor. — Ed.

Let $\mathcal{F}$ be a family of exactly f basic sets, closed under union. The UCS conjecture is trivial if $f \leq 1$, so we require that $f \geq 2$. Union-closure means that for all $S, T \in \mathcal{F}$ we have $S \cup T \in \mathcal{F}$. Once again, we write $L = L(\mathcal{F}) = \bigcup_{S \in \mathcal{F}} S$ for the union of all sets in $\mathcal{F}$, putting $\ell = \ell(\mathcal{F}) = |L|$ whenever L happens to be finite. For each $a \in L$, let us also define a new family of sets,

$$\mathcal{F}_a := \{S \in \mathcal{F} : a \in S\},$$

consisting of those sets of $\mathcal{F}$ containing a . It is then natural to write $f_a := |\mathcal{F}_a|$. As well, for each positive integer i , we let g_i denote the number of sets in $\mathcal{F}$ which contain exactly i ur-elements. That is,

$$g_i := |\{A \in \mathcal{F} : |A| = i\}|.$$

We will be studying families $\mathcal{F}$ for which L is finite. A justification (due to Poonen [1]) of this apparent loss of generality will be examined shortly.

Finally, we define the *density* of $\mathcal{F}$ by

$$d(\mathcal{F}) := \frac{1}{\ell f} \sum_{a \in L} f_a = \frac{1}{\ell f} \sum_{S \in \mathcal{F}} |S|.$$

The two representations of $d(\mathcal{F})$ given above are equivalent, since both sums can be interpreted as counting each set of $\mathcal{F}$ as many times as its cardinality. More formally, both sums count the ordered pairs (a, S) with $a \in S \in \mathcal{F}$. The first sum counts this collection by letting a run through L , and for each a counting the f_a basic sets of $\mathcal{F}$ to which a belongs. The second sum lets S run through $\mathcal{F}$, and for each S counts the $|S|$ ur-elements that belong to it. The density of $\mathcal{F}$ can be interpreted in two useful ways:

1. as the ratio of the average f_a to the size f of $\mathcal{F}$,
2. as the ratio of the average set size $|S|$ to the maximum set size ℓ .

Associated with the density concept is the quantity

$$s_n := \min_{\ell(\mathcal{F})=n} d(\mathcal{F}),$$

that is, the minimum possible density of a union-closed family $\mathcal{F}$ for which $\ell = n$. This is defined for all positive integers n .

3. PRELIMINARIES. For the sake of introducing the reader to the general methods of this subject, we will begin by presenting the proofs of two previously established theorems. The first, due to Poonen [1], will allow us to restrict our attention to families of whose sets are all finite, or equivalently, families for which L is finite.

Theorem (Poonen). With respect to bounds on f_a for some $a \in L$, it is sufficient to consider families $\mathcal{F}$ for which L is finite.

Proof. Let $\mathcal{B}$ be the collection of all equivalence classes of the following relation, defined on the elements of L : $a \sim b$ if and only if $\mathcal{F}_a = \mathcal{F}_b$. Since each $B \in \mathcal{B}$ consists of elements which always appear together in the sets of $\mathcal{F}$, every $B \in \mathcal{B}$ is of the form $S_1 \cap S_2 \cap \dots \cap S_r$ where $S_1, S_2, \dots, S_r$ are the sets of $\mathcal{F}$ in which the elements of B appear. The number of such intersections is at most the number of subsets of $\mathcal{F}$, since forming an intersection of this kind required first selecting a subset $\{S_1, S_2, \dots, S_r\} \subset \mathcal{F}$. The number of subsets of $\mathcal{F}$ is 2^f , so we have $|\mathcal{B}| \leq 2^f$. In other words, $\mathcal{B}$ is always finite. Next, for each $B \in \mathcal{B}$, let us replace every occurrence of an element of B in the sets of $\mathcal{F}$ with some fixed $a \in B$, thereby producing a new family $\mathcal{F}'$. For example, if $\mathcal{F} = \{\{1, 2, 3\}, \{1, 2, 4, 5\}, \{1, 2, 3, 4, 5\}\}$, then $\mathcal{B} = \{\{1, 2\}, \{3\}, \{4, 5\}\}$ and we may take $\mathcal{F}' = \{\{1, 3\}, \{1, 4\}, \{1, 3, 4\}\}$. Let $L' := L(\mathcal{F}') = \bigcup_{S \in \mathcal{F}'} S$ and $\ell' := |L'|$. It is clear that $\ell' = |\mathcal{B}|$ (and hence that L' is finite) since L' contains exactly one representative member from each $B \in \mathcal{B}$, and nothing else. The construction $\mathcal{F}'_b = \{S \in \mathcal{F}' : b \in S\}$ applies for all $b \in L'$. The relation between $\mathcal{F}_b$ and $\mathcal{F}'_b$ for a given $b \in L'$ is that $f'_b = |\mathcal{F}'_b| = |\mathcal{F}_b| = f_b$. This is because our compression of the elements equivalent to b into b itself has no effect on the number of distinct sets in $\mathcal{F}_b$. Finally, for any particular $a \in L$, there exists a $b \in L'$ such that $f_a = f'_b$. This b is simply the unique element of the equivalence class $[a] = \{x : x \sim a\}$ that was selected to belong to L' . So any bound for f'_b dependent on L' being finite applies also to f_a , where L need not be finite. $\square$

The next theorem, due to Wójcik [2], plays an important role in the proof of our main result. Recall that s_n is the minimum possible density of a union-closed family $\mathcal{F}$ for which $\ell(\mathcal{F}) = n$.

Theorem (Wójcik). For any given $S \in \mathcal{F}$ with $|S| = k > 0$, we have

$$\frac{1}{k} \sum_{a \in S} f_a \geq s_k f.$$

Proof. Select an $S \in \mathcal{F}$ with $|S| = k > 0$. Let Γ be the set of equivalence classes of the following relation defined for all $U, V \in \mathcal{F}$: $U \sim V$ if and only if $U - S = V - S$. That is, U and V are equivalent if they differ only inside S . For each equivalence class $\mathcal{P} \in \Gamma$, define $\mathcal{P}' := \{U \cap S : U \in \mathcal{P}\}$. Then $|\mathcal{P}| = |\mathcal{P}'|$. The reason is that certainly $U = V$ implies $U \cap S = V \cap S$, and when $U \sim V$ the reverse implication is also true, since

$$U = (U \cap S) \cup (U - S) = (V \cap S) \cup (V - S) = V.$$

Since Γ is a partition of $\mathcal{F}$ (as with any equivalence relation),

$$\sum_{\mathcal{P} \in \Gamma} |\mathcal{P}| = f.$$

Notice that whenever $U \sim V$, we have $U - S = V - S = (U \cup V) - S$, because of the set-theoretic fact that $X = Y$ implies $X = X \cup Y$. It follows that for any

given equivalence class $\mathcal{P} \in \Gamma$, whenever $U, V \in \mathcal{P}$, we also have $U \cup V \in \mathcal{P}$. That is, each $\mathcal{P}$ is union-closed. As an easy set-theoretic consequence, each $\mathcal{P}'$ is also union-closed [in view of the law $(U \cap S) \cup (V \cap S) = (U \cup V) \cap S$]. For any $U \in \mathcal{P}$, observe that $U \cup S \in \mathcal{F}$, because $U, S \in \mathcal{F}$ and $\mathcal{F}$ is union-closed. Since trivially $U - S = (U \cup S) - S$, we find also that $U \cup S \sim U$, or what is the same, $U \cup S \in \mathcal{P}$. This gives $S = (U \cup S) \cap S \in \mathcal{P}'$. Now all the sets in $\mathcal{P}'$ are subsets of S , so certainly $\bigcup_{X \in \mathcal{P}'} X \subset S$. The reverse inclusion is also true, since for any $a \in S$, we may take $X = S$ to get $a \in X \in \mathcal{P}'$ and hence $a \in \bigcup_{X \in \mathcal{P}'} X$. It is now clear that $\bigcup_{X \in \mathcal{P}'} X = S$. Consequently, $\ell(\mathcal{P}') = |S| = k$, and since s_k is a minimum taken over union-closed families with $\ell = k$, we conclude that $d(\mathcal{P}') \geq s_k$. Recalling that $|\mathcal{P}'| = |\mathcal{P}|$, this leads to the inequality

$$\alpha := \sum_{\mathcal{P} \in \Gamma} d(\mathcal{P}') |\mathcal{P}'| \geq \sum_{\mathcal{P} \in \Gamma} s_k |\mathcal{P}| = s_k \sum_{\mathcal{P} \in \Gamma} |\mathcal{P}| = s_k f. \quad (*)$$

From the definition of density,

$$d(\mathcal{P}') = \frac{1}{\ell(\mathcal{P}') |\mathcal{P}'|} \sum_{X \in \mathcal{P}'} |X|.$$

Multiplying through by $|\mathcal{P}'|$ and putting $\ell(\mathcal{P}') = k$ lets us rewrite α as

$$\alpha = \frac{1}{k} \sum_{\mathcal{P} \in \Gamma} \sum_{X \in \mathcal{P}'} |X| = \frac{1}{k} \sum_{\mathcal{P} \in \Gamma} \sum_{U \in \mathcal{P}} |U \cap S|.$$

Since Γ partitions $\mathcal{F}$, each $U \in \mathcal{F}$ belongs to precisely one $\mathcal{P} \in \Gamma$. So a double summation over all $U \in \mathcal{P}$ and $\mathcal{P} \in \Gamma$ is equivalent to a single summation over all $U \in \mathcal{F}$. That is,

$$\alpha = \frac{1}{k} \sum_{U \in \mathcal{F}} |U \cap S|.$$

Finally, consider the last sum. As U runs through $\mathcal{F}$, it counts the $|U \cap S|$ ur-elements in S that belong to U . Instead, we can let a run through S , and count the f_a basic sets in $\mathcal{F}$ to which a belongs. This leads to

$$\alpha = \frac{1}{k} \sum_{a \in S} f_a.$$

Putting this into $(*)$, the result $\alpha \geq s_k f$ is the required theorem. $\square$

The following lemma establishes a lower bound on the minimum density s_n . This bound appears without proof in [2].

Lemma. For $n \geq 2$, we have $s_n \geq \frac{2}{n+2}$.

Proof. Consider $\mathcal{F}$ with $\ell = n \geq 2$. Recall that g_i is the number of sets in $\mathcal{F}$ containing exactly i ur-elements. Since L is finite, $L \in \mathcal{F}$, and of course

$|L| = \ell = n$. Moreover, no other set of $\mathcal{F}$ has n members, since each is a subset of L . It follows that $g_n = 1$. Also, $g_0 \leq 1$, since the empty set $\emptyset$ either belongs to $\mathcal{F}$ ($g_0 = 1$), or does not ($g_0 = 0$). As well, we have $g_1 \leq n$, since there can only be as many one-element sets in $\mathcal{F}$ as there are elements in L . Using the hypothesis $n \geq 2$ to note that $2 - n \leq 0$, we now have

$$n^2 g_n - 2n g_0 + (2 - n) g_1 \geq n^2(1) - 2n(1) + (2 - n)(n) = 0.$$

Next, note that $(n + 2)i - 2n \geq 0$ for all $i \geq 2$. Of course, all g_i are nonnegative, since they count something. Thus by adding only nonnegative terms to the preceding inequality, we get

$$\beta := \sum_{i=2}^{n-1} ((n + 2)i - 2n) g_i + n^2 g_n - 2n g_0 + (2 - n) g_1 \geq 0.$$

In the above summation sign, the ‘missing’ terms $((n + 2)i - 2n) g_i$ for $i = 0, 1, n$ are respectively $-2n g_0$, $(2 - n) g_1$, and $n^2 g_n$, which all appear following the summation sign. So β can be rewritten as

$$\beta = \sum_{i=0}^n ((n + 2)i - 2n) g_i.$$

We now simplify β further, as follows:

$$\begin{aligned} \beta &= \sum_{i=0}^n ((n + 2)i - 2n) g_i \\ &= (n + 2) \sum_{i=0}^n i g_i - 2n \sum_{i=0}^n g_i \\ &= (n + 2) \sum_{S \in \mathcal{F}} |S| - 2n f. \end{aligned}$$

In the last step, we used the following facts:

$$\sum_{i=0}^n i g_i = \sum_{S \in \mathcal{F}} |S|, \quad \text{and} \quad \sum_{i=0}^n g_i = f.$$

The second of these is obvious, since the f sets of $\mathcal{F}$ are partitioned into those that contain 0 elements, those that contain 1 element, etc. The first fact is true for the following reason. When we sum $|S|$ over all $S \in \mathcal{F}$, we are counting the ordered pairs (a, S) with $a \in S \in \mathcal{F}$, as discussed earlier. These same pairs may be partitioned into collections of sizes $0g_0, 1g_1, 2g_2, \dots, ng_n$, in this way: for each $i \in \{0, 1, \dots, n\}$, group together those pairs (a, S) for which $|S| = i$. The number of such pairs is $i g_i$, since we have g_i choices for a set $S \in \mathcal{F}$ with $|S| = i$, followed by i choices for an $a \in S$. Using the fact that $\beta \geq 0$, it now follows easily that

$$d(\mathcal{F}) = \frac{1}{fn} \sum_{S \in \mathcal{F}} |S| \geq \frac{2}{n + 2}.$$

Since this holds for *every* $\mathcal{F}$ with $\ell(\mathcal{F}) = n$, we see that $s_n \geq 2/(n + 2)$. $\square$

4. MAIN RESULT. We are finally in a position to prove our main result, which provides a lower bound on the number of sets of $\mathcal{F}$ to which some single element of L must belong. The UCS conjecture posits a lower bound of $f/2$. We give a weaker lower bound which depends on both f and ℓ , provided $\ell \geq 4$. That our bound is weaker is due to the fact that

$$\frac{2f}{1 + \sqrt{2\ell + 1}} \leq \frac{f}{2} \quad (\ell \geq 4).$$

This is easily verified by clearing denominators, isolating the square root, and squaring both sides. For ease of notation in the proof, we write

$$\lambda(f, \ell) := \frac{2f}{1 + \sqrt{2\ell + 1}}.$$

Theorem. For any $\mathcal{F}$ with $\ell \geq 4$, there is an element $a \in L$ such that

$$f_a \geq \frac{2f}{1 + \sqrt{2\ell + 1}}.$$

Proof. Let $\mathcal{F}$ have $\ell = n \geq 4$. We begin by establishing two lower bounds on the maximum f_a , taken over all $a \in L$. Let S be a nonempty set of minimal size in $\mathcal{F}$, and let this minimal size be denoted by k . Now the average set size in $\mathcal{F}$ is certainly at least as great as the minimum positive set size. Hence if $k > n/2$, the average set size is $> k/2$. Observe however that by the two equivalent ways of counting pairs (a, S) with $a \in S \in \mathcal{F}$, we then have

$$\frac{1}{f} \sum_{U \in \mathcal{F}} |U| > \frac{n}{2} \quad \Rightarrow \quad \frac{1}{n} \sum_{a \in L} f_a > \frac{f}{2}.$$

This says that the average f_a is $> f/2$. This can only occur if some particular f_a is $> f/2$. For this particular a , certainly $f_a \geq \lambda(f, n)$, so we are done. The remaining case involves $k \leq n/2$. If $k = 1$, then S is a singleton set. Take a to be the unique member of S . Define $\mathcal{F}_{(a)} := \mathcal{F} - \mathcal{F}_a$ [i.e., the family of all sets of $\mathcal{F}$ not containing a], and $f_{(a)} := |\mathcal{F}_{(a)}|$. Notice that $f_a \geq f_{(a)}$, since for every $U \in \mathcal{F}_{(a)}$, there exists a set $\tilde{U} = U \cup \{a\} \in \mathcal{F}_a$, with different U 's leading to different $\tilde{U}$'s. Since $\{\mathcal{F}_a, \mathcal{F}_{(a)}\}$ is a partition of $\mathcal{F}$, we have $f_a + f_{(a)} = f$, and since $f_a \geq f_{(a)}$, it follows that $f_a \geq f/2$. Again, this obviously gives $f_a \geq \lambda(f, n)$, and we are done. It remains to consider the case $k \geq 2$. By the theorem of Wójcik, we may start with the fact that

$$\frac{1}{k} \sum_{a \in S} f_a \geq s_k f.$$

Replacing s_k with the lower bound for s_k derived in the lemma, we get

$$\frac{1}{k} \sum_{a \in S} f_a \geq \frac{2f}{k + 2}.$$

Since the maximum f_a is at least as great as the average f_a , the preceding formula produces our first bound:

$$\max_{a \in L} f_a \geq \frac{2f}{k+2}. \quad (\dagger)$$

Next, notice that since S is a minimal nonempty set of $\mathcal{F}$, we have $g_i = 0$ for all $i \in \{1, 2, \dots, k-1\}$. It clearly follows that

$$\sum_{i=1}^{n-1} (i-k)g_i \geq 0.$$

In this summation, we wish to restore the ‘missing’ terms corresponding to $i = 0, n$, which would be $(n-k)g_n - kg_0$, without changing the fact that the sum is nonnegative. This can be done, since

$$(n-k)g_n - kg_0 \geq (n-k)(1) - k(1) = n-2k \geq 0$$

[recall that $k \leq n/2$]. As a result,

$$\gamma := \sum_{i=0}^n (i-k)g_i \geq 0.$$

We simplify this expression for γ as follows:

$$\gamma = \sum_{i=0}^n ig_i - k \sum_{i=0}^n g_i = \sum_{U \in \mathcal{F}} |U| - kf = \sum_{a \in L} f_a - kf,$$

each step of which has been discussed previously. With this in place, $\gamma \geq 0$ becomes

$$\frac{1}{n} \sum_{a \in L} f_a \geq \frac{kf}{n}.$$

This gives us our second bound:

$$\max_{a \in L} f_a \geq \frac{kf}{n}. \quad (\dagger\dagger)$$

Finally, we proceed by contradiction. Assume that the maximum f_a is $< \lambda(f, n)$. Then $(\dagger)$ entails that

$$\frac{2f}{k+2} < \lambda(f, n) = \frac{2f}{1 + \sqrt{2n+1}},$$

from which it is easy to deduce that $k > \sqrt{2n+1} - 1$. On the other hand, $(\dagger\dagger)$ entails that

$$\frac{kf}{n} < \lambda(f, n) = \frac{2f}{1 + \sqrt{2n+1}},$$

and accordingly

$$k > \frac{2n}{1 + \sqrt{2n+1}}.$$

The two bounds on k imply that

$$\frac{2n}{1 + \sqrt{2n+1}} < \sqrt{2n+1} - 1.$$

Multiplying this out yields the contradiction $2n < 2n$. The proof is complete once the general variable ℓ is put back for its specific value n . $\square$

5. FURTHER RESEARCH. Wójcik has conjectured that the union-closed family $\mathcal{F}_0$ with $\ell = n$ which minimizes the density $d(\mathcal{F})$ has one of the forms

$$\mathcal{F}_0 = P(\lceil \log_2 n \rceil) \cup \{L\} \quad \text{or} \quad \mathcal{F}_0 = P(\lfloor \log_2 n \rfloor) \cup \{L\},$$

where $P(x)$ is the power set (i.e., the set of all subsets) of some $X \subseteq L$ such that $|X| = x$. This would mean that

$$s_n \approx \frac{\log_2 n}{2n}.$$

Such would be a sharper bound on s_n than the quantity $2/(n+2)$ used above. If this turns out to be true, it would be possible to use the method of our main theorem to obtain a correspondingly sharper lower bound on $\max_{a \in L} f_a$, and thereby to approach a little more closely to the UCS conjecture.

There is ongoing work which aims to prove that

$$s_n \geq \frac{\log_2 n}{2n}.$$

This is where the matter rests currently.

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Three Impossible Construction Problems of Antiquity

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1. INTRODUCTION. The two oldest planimetric tools known to history are the unmarked straightedge and compass. Their practical applications in architecture and surveying dates back to remote antiquity, and there has been a mathematical theory of straightedge and compass constructions since well before the time of Euclid. Although the ancients were very clever and discovered a great many constructions, there were three simple problems which stubbornly resisted their best efforts. Word of these challenging problems spread rapidly, and they soon became a famous part of the mathematical folklore. They were:

1. *To double the cube.* Given a cube, construct one edge of another cube having twice the volume of the first.
2. *To trisect the angle.* Given an angle, construct two rays in its interior so as to subdivide it into three congruent parts.
3. *To square the circle.* Given a circle, construct a square whose area is equal to that of the circle.

It will perhaps come as a disappointment that all three problems have been settled in the negative: none of the constructions can be done. This statement may seem rather mundane on its face, but when one stops to mull it over, it is actually quite remarkable. No matter how ingenious we are, no matter how much time we spend, and no matter what inspired maneuvers we try, we will never hit on a solution. But how can something like this be proved?

A promising new pathway opened in the Seventeenth Century, when René Descartes introduced the world to the idea of *analytic geometry*. For the first time, algebraic reasoning was brought to bear on geometry. At the same time, algebra itself grew rapidly and reached new levels of abstraction. By the late Eighteenth Century, the great Carl Friedrich Gauss understood how to apply abstract algebra to the question of constructibility, and did some important work on the constructibility of regular polygons. The negative resolution of the first two ancient problems was secured a few decades later by Pierre Wantzel. The third problem — squaring the circle — would hold out another fifty years before it too fell. With his work of 1882 on π , Ferdinand von Lindemann brought a close to this long and fascinating chapter in the development of mathematics.

[†]The staff of MATH SURVEY are grateful to Mr Stern for agreeing to share this article, which was prepared as supplemental reading for his Honors Geometry course. — Ed.

In this article we study the negative resolutions of the three ancient problems. Our aim is to give the reader some feeling for the beautiful interplay between algebraic and geometric ideas that characterizes the whole subject.

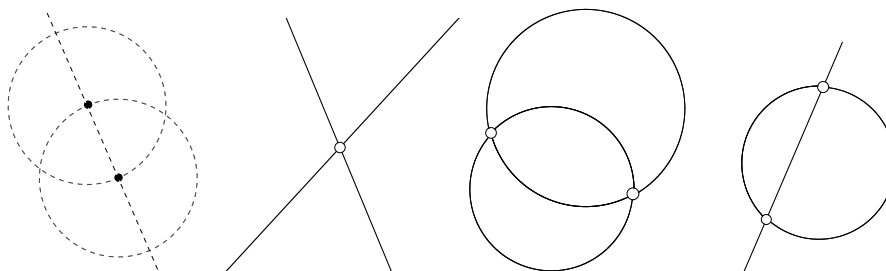
2. CONSTRUCTIBILITY. To develop a theory of constructibility, our first task must be to take the problem out of the physical realm and bring it into the domain of formal mathematics. We must decide what aspects of physical straightedges and compasses are really relevant, and then simulate them within the confines of formal geometry. The first thing to notice is that we lose nothing by thinking of our straightedge as being infinitely long, or our compass as being capable of an arbitrarily large radius. Although these are not physically realistic traits, any particular figure that can be constructed in finitely many steps will certainly involve some maximal distance.

With this in mind, we define a property of *constructibility*. The only things capable of having this property are points, lines, and circles. In fact, let us co-opt the word ‘object’ as a blanket term to include points, lines, and circles. Objects will be represented by the Greek letter omega: capital ‘ Ω ’, and lowercase ‘ ω ’. Throughout this article, the symbol $(A)_B$ denotes the circle with center A that contains B , formally $\{X : AX = AB\}$.

Definition 1. *Certain objects are said to be descendants of others (figure 1).*

1. *The descendants of a pair $\{A, B\}$ of distinct points are $\overleftrightarrow{AB}$, $(A)_B$, $(B)_A$.*
2. *The descendant of a pair $\{\ell_1, \ell_2\}$ of distinct intersecting lines is the unique point in $\ell_1 \cap \ell_2$.*
3. *The descendants of a pair $\{c_1, c_2\}$ of distinct intersecting circles are the points of $c_1 \cap c_2$.*
4. *The descendants of a pair $\{\ell, c\}$, comprised of a line and a circle which intersect, are the points of $\ell \cap c$.*

Figure 1: Relations of descent among objects. Descendant lines and circles are shown with dashes, while descendant points are shown as small open discs.



Definition 2. An object Ω is said to be constructible if there exists a sequence $\langle \omega_0, \omega_1, \omega_2, \dots, \omega_N \rangle$ such that:

1. $\omega_0 = (0, 0)$ and $\omega_1 = (1, 0)$
2. $\omega_N = \Omega$
3. for each $i \geq 2$, ω_i is a descendant of some pair $\{\omega_j, \omega_k\}$ with $j < k < i$.

In addition, $(0, 0)$ and $(1, 0)$ are constructible. The set of all constructible objects is denoted by $\mathcal{C}$.

The relation of descent among objects is intended to mimic the process of drawing with a straightedge and a compass. For example, the descent of $\overline{AB}$ from $\{A, B\}$ should suggest the process of positioning a straightedge through two previously marked locations and tracing a line along its edge. Likewise, the descent of a point $P \in \ell_1 \cap \ell_2$ from $\{\ell_1, \ell_2\}$ suggests the process of placing a mark at the juncture of two previously drawn lines. By definition, an object Ω is constructible if we can begin with the standard point pair $\{(0, 0), (1, 0)\}$, find one of its descendants, then a descendant of the objects found so far, and so on, repeating this until we reach Ω . The sequence of objects $\langle \omega_0, \omega_1, \omega_2, \dots, \omega_N \rangle$ is a mathematical model of the underlying sequence of drawing operations that produces Ω from a given pair of reference points. The reason for $(0, 0)$ and $(1, 0)$ as reference points is to establish a fixed unit of distance.

It is important to avoid the false notion that once a line or a circle has been constructed, so have all the points on it. A point has been constructed only once it has been shown to lie on *two* constructible objects: a pair of lines, a pair of circles, or a line-circle pair. The objects one proposes to use for that purpose must first be proven *distinct* and *intersecting*. In practice, this step is frequently omitted. Instead, typically a diagram is shown which offers visual evidence that all required intersections actually occur, as well as guidance on how to supply a formal proof if necessary. With a few instructive exceptions, this is the approach we will take below.

In case we require a formal proof that two particular objects intersect, the following principles of elementary geometry are useful:

1. The lines ℓ_1 and ℓ_2 intersect if some transversal ℓ forms unequal angles with them.
2. If a line ℓ passes through an interior point of the circle c , then ℓ and c have two intersection points.
3. Let c_1 and c_2 be circles of radii r_1 and r_2 , with centers d units apart. Then c_1 and c_2 have two intersection points if $|r_1 - r_2| < d < r_1 + r_2$.

A useful application of the third principle in constructibility theory is the idea that $(A)_B$ and $(B)_A$ *must have two intersection points*. The reason is that in these circles, $r_1 = r_2 = d$, and certainly $|d - d| < d < d + d$.

We close this section by demonstrating the constructibility of the coordinate axes ℓ_x and ℓ_y , as well as the points $(\pm 1, 0)$ and $(0, \pm 1)$. It will be useful to introduce the notations

$$O = (0, 0), \quad I = (1, 0), \quad I' = (-1, 0), \quad J = (0, 1), \quad J' = (0, -1),$$

as well as

$$c^* = \{(x, y) : x^2 + y^2 = 1\}$$

for the *standard unit circle*, otherwise denoted by $(O)_I$. That I is constructible was part of Definition 2. To see that the x -axis is constructible, note that the sequence $\langle O, I, \ell_x \rangle$ trivially satisfies the requirements of the definition. Next, consider the sequence

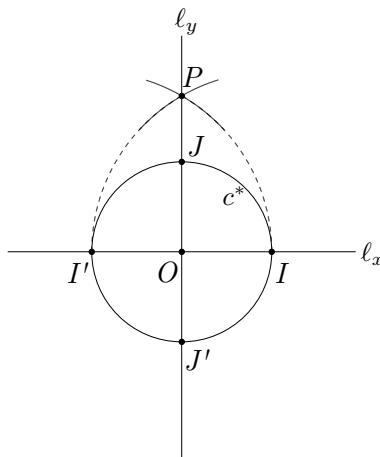
$$\langle O, I, \ell_x, c^*, I' \rangle.$$

The validity of this sequence is obvious once we note that I' is a descendant of the pair $\{\ell_x, c^*\}$ (figure 2). As for the y -axis, figure 2 illustrates the sequence

$$\langle O, I, \ell_x, c^*, I', (I)_{I'}, (I')_I, P, \ell_y \rangle,$$

where $P = (0, \sqrt{3})$. Notice that the circles $(I)_{I'}$ and $(I')_I$ are only partially drawn, in order to minimize clutter in the diagram.

Figure 2: Construction of the y -axis.



The validity of this sequence is obvious up to P . The equations of the circles $(I)_{I'}$ and $(I')_I$ are $(x - 1)^2 + y^2 = 4$ and $(x + 1)^2 + y^2 = 4$. Eliminating $4 - y^2$ between these equations, we find that $x = 0$ and $y^2 = 3$, yielding the descendant $P = (0, \sqrt{3})$. To complete the sequence, note that ℓ_y descends from $\{O, P\}$. The points J and J' may be constructed immediately as well, since each of these points descends from the pair $\{c^*, \ell_y\}$.

3. CLASSICAL CONSTRUCTIONS. In this section we briefly survey some important classical constructions. There are two good reasons for reviewing these here. First, such constructions are interesting and elegant in their own right. Every student of the subject ought to have seen them. Second, they will play a critical role in the translation process that turns constructibility problems into questions of abstract algebra. The style here will be much less formal than in the previous section. Instead of exhibiting a valid sequence $\langle O, I, \dots, \Omega \rangle$, we demonstrate the constructibility of an object Ω by presenting a diagram that indicates all the appropriate steps. In order to aid the reader in interpreting a static diagram as an unfolding construction process, all the objects will be numbered in their order of appearance in the formal sequence. First, it will be useful to sum up the results derived in the previous section.

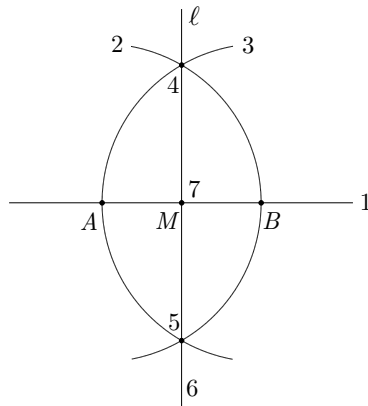
Theorem 1 (Basic Objects). *The coordinate axes ℓ_x and ℓ_y , the unit circle c^* , and the points I, J, I', J' all belong to the set $\mathcal{C}$ of constructible objects.*

Suppose we are given two constructible points A and B . It is natural to ask for a construction of their midpoint M , as well as their perpendicular bisector ℓ . Figure 3 indicates ‘relative constructions’ of these objects. This means that M or ℓ can be constructed once we know a pair of constructions for A and B . The diagram supplies two sequences $\langle A, B, \dots, M \rangle$ and $\langle A, B, \dots, \ell \rangle$, which satisfy clause 3 of Definition 2. By hypothesis, there are valid sequences $\langle O, I, \dots, A \rangle$ and $\langle O, I, \dots, B \rangle$, so it is clear how to string together a valid sequence for M or ℓ . All constructions described henceforward will be ‘relative’ in this sense.

Theorem 2 (Midpoint, Perpendicular Bisector). *Given distinct points A and B in $\mathcal{C}$, their midpoint M and perpendicular bisector ℓ also belong to $\mathcal{C}$.*

Proof. See figure 3. □

Figure 3: Construction of the midpoint and perpendicular bisector.

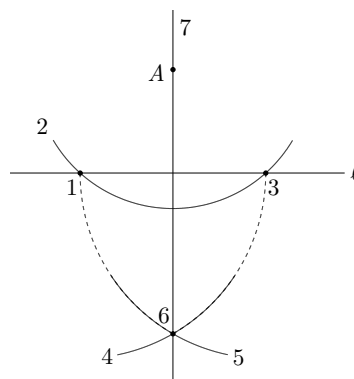


Next we construct the perpendicular of a given line through a given point.

Theorem 3 (Perpendiculars). *Given a point $A \in \mathcal{C}$ and a line $\ell \in \mathcal{C}$, the perpendicular of ℓ through A belongs to $\mathcal{C}$.*

Proof. See figure 4. The point labeled ‘1’ can be taken as the intersection of ℓ with one of the coordinate axes if necessary. By Theorem 1, this is descended from a pair of constructible lines, and is therefore also constructible. $\square$

Figure 4: Construction of perpendiculars.

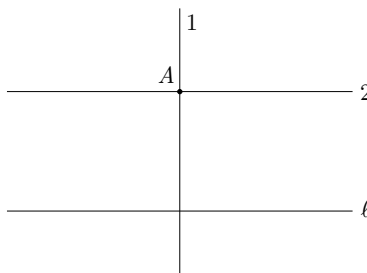


Certain details of the next construction may safely be hidden from view, since they are merely applications of previously proven theorems. Thus the lines labeled ‘1’ and ‘2’ in figure 5 are not really constructed in one step each, but require the technique of Theorem 3. Such partial occlusions of previously studied techniques will be used without comment from now on.

Theorem 4 (Parallels). *Given a point $A \in \mathcal{C}$ and a line $\ell \in \mathcal{C}$, the parallel of ℓ through A belongs to $\mathcal{C}$.*

Proof. See figure 5. Construct the perpendicular of ℓ through A , followed by the perpendicular of line ‘1’ through A . This yields the required parallel of ℓ . $\square$

Figure 5: Construction of parallels.



Theorem 5 (Copying a Segment). *Given three points $A, B, C \in \mathcal{C}$, and a constructible line ℓ through C , $\mathcal{C}$ contains a point $X \in \ell$ such that $CX = AB$.*

Proof. See figure 6. Line ‘2’ is parallel to ℓ , and line ‘5’ is parallel to $\overleftrightarrow{AC}$. $\square$

Theorem 6 (Angle Bisector). *Given noncollinear points A, B, C in $\mathcal{C}$, the bisector of $\angle ABC$ belongs to $\mathcal{C}$.*

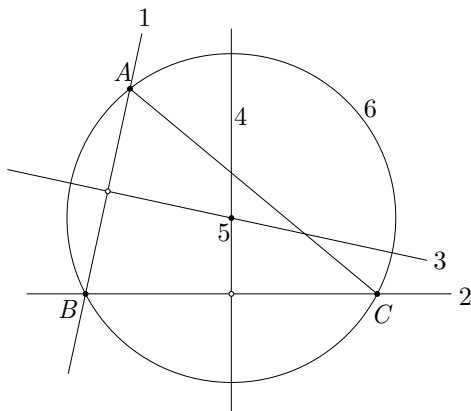
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Theorem 7 (Circumcenter, Circumcircle). *Given noncollinear points A, B, C in $\mathcal{C}$, the circumcenter and circumcircle of $\triangle ABC$ belong to $\mathcal{C}$.*

Proof. See figure 8. □

Figure 8: Construction of the circumcenter and circumcircle.



Finally, we leave the following two theorems as a challenging exercise for the reader to prove. These theorems will not be used in what follows.

Theorem 8 (Copying an Angle). *Given noncollinear points $A, B, C \in \mathcal{C}$, a point $E \in \mathcal{C}$, and a constructible line ℓ through E , $\mathcal{C}$ contains a pair of points D, F with $D \in \ell$ and $\angle DEF \cong \angle ABC$.*

Theorem 9 (Subdividing a Segment). *Given distinct points $A, B \in \mathcal{C}$ and an integer $n > 1$, $\mathcal{C}$ contains points $P_1, P_2, \dots, P_{n-1}$ between A and B which subdivide AB into n equal parts: $AP_1 = P_1P_2 = \dots = P_{n-1}B$.*

Theorems 6 and 9 produce an expectation that angle trisection ought to be possible. They will make the negative result seem all the more surprising.

4. FIELDS AND QUADRATIC EXTENSIONS. In order to establish the possibility of a construction, we must only find a single way to accomplish it. But what if we suspect that the task is impossible? It is not feasible to consider every proposed solution and rule them out one by one. Indeed, if our viewpoint remains confined to geometry, it is difficult to see how an impossibility claim could ever be conclusively demonstrated. This is why the negative results on doubling the cube and trisecting the angle went unsuspected for thousands of years. The answer, it turns out, is to radically change our viewpoint, translating the entire issue into the language of algebra.

The kind of algebra to which we are referring is not the kind one learns in grade school. It is ‘abstract algebra’, meaning the study of mathematical systems in which some operations obey particular sets of laws. Different sets of laws define different kinds of mathematical system. However, in this article we will be exclusively concerned with mathematical systems known as *fields*.

Definition 3. A field $\mathbb{F}$ is a set of numbers with the following properties:

1. $0, 1 \in \mathbb{F}$.
2. If $a, b \in \mathbb{F}$, then $a + b \in \mathbb{F}$.
3. If $a, b \in \mathbb{F}$, then $a - b \in \mathbb{F}$.
4. If $a, b \in \mathbb{F}$, then $ab \in \mathbb{F}$.
5. If $a, b \in \mathbb{F}$ and $b \neq 0$, then $a/b \in \mathbb{F}$.

The four operations mentioned here — addition, subtraction, multiplication, division — are called the *rational operations*. A field is just a set of numbers that contains $\{0, 1\}$ and is closed under each rational operation. Observe that closure under division may be replaced with closure under *reciprocation*:

- 5'. If $a \in \mathbb{F}$ and $a \neq 0$, then $a^{-1} \in \mathbb{F}$.

The reason is that division can be viewed as a byproduct of multiplication and reciprocation: $a/b = a \cdot b^{-1}$.

The three standard examples of fields that one encounters in school are:

1. $\mathbb{Q}$, the set of *rational numbers*.
2. $\mathbb{R}$, the set of *real numbers*.
3. $\mathbb{C}$, the set of *complex numbers* ($a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$).

In our studies we will have no reason to consider complex numbers, so let us agree to restrict our attention from this point on to *real fields*, that is, fields $\mathbb{F}$ such that $\mathbb{F} \subset \mathbb{R}$. We will use the term ‘field’ synonymously with ‘real field’.

Theorem 10. For every field $\mathbb{F}$, we have $\mathbb{Q} \subset \mathbb{F}$.

Proof. We must only show that $\mathbb{Z} \subset \mathbb{F}$, where $\mathbb{Z}$ is the set of integers. Once all the integers belong to $\mathbb{F}$, so do all their quotients by closure under division. Now $0, 1 \in \mathbb{F}$. If we assume for induction that $n \in \mathbb{F}$ where n is a positive integer, then by closure under addition, $n + 1 \in \mathbb{F}$. By induction, every positive integer belongs to $\mathbb{F}$. If n is a negative integer, we also have $n = 0 - |n| \in \mathbb{F}$, by closure under subtraction. Thus $\mathbb{Z} \subset \mathbb{F}$, and hence $\mathbb{Q} \subset \mathbb{F}$. $\square$

A field $\mathbb{F}$ is said to be *Euclidean* if $\sqrt{a} \in \mathbb{F}$ for every positive number $a \in \mathbb{F}$. A simple example of a Euclidean field is $\mathbb{R}$, and a simple non-example is $\mathbb{Q}$. As we will see, although 2 is positive and rational, $\sqrt{2}$ is not a rational number.

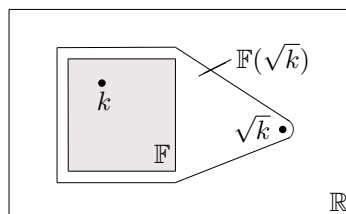
If a field $\mathbb{F}$ is contained in another field $\mathbb{G}$, we say that $\mathbb{G}$ is an *extension* of $\mathbb{F}$. The following theorem introduces a process for generating simple examples of extensions. These are called *quadratic extensions*. A schematic diagram of a quadratic extension is shown in figure 9 below.

Theorem 11 (Quadratic Extension Theorem). *Let $\mathbb{F}$ be a field, and suppose $k > 0$ is such that $k \in \mathbb{F}$ but $\sqrt{k} \notin \mathbb{F}$. Define*

$$\mathbb{F}(\sqrt{k}) = \{a + b\sqrt{k} : a, b \in \mathbb{F}\}. \quad (1)$$

Then $\mathbb{F}(\sqrt{k})$ is a field, and is in fact the smallest extension of $\mathbb{F}$ containing $\sqrt{k}$.

Figure 9: Schematic representation of a quadratic extension.



Proof. First, $\mathbb{F}(\sqrt{k}) \subset \mathbb{R}$, since its elements are $a + b\sqrt{k}$, where $a, b, \sqrt{k} \in \mathbb{R}$. Clearly $\mathbb{F} \subset \mathbb{F}(\sqrt{k})$, since every $a \in \mathbb{F}$ can be viewed as $a + 0\sqrt{k}$. Since $0, 1$ belong to $\mathbb{F}$, they automatically belong to $\mathbb{F}(\sqrt{k})$. Also, $\sqrt{k} \in \mathbb{F}(\sqrt{k})$, as $\sqrt{k} = 0 + 1\sqrt{k}$. To see that $\mathbb{F}(\sqrt{k})$ is closed under the rational operations, consider the following computations:

1. $(a + b\sqrt{k}) \pm (c + d\sqrt{k}) = (a \pm c) + (b \pm d)\sqrt{k}$.
2. $(a + b\sqrt{k})(c + d\sqrt{k}) = (ac + bdk) + (ad + bc)\sqrt{k}$.
3. $\frac{1}{a + b\sqrt{k}} = \frac{a - b\sqrt{k}}{(a + b\sqrt{k})(a - b\sqrt{k})} = \left(\frac{a}{a^2 - b^2k}\right) + \left(\frac{-b}{a^2 - b^2k}\right)\sqrt{k}$.

In each computation, $a, b, c, d, k \in \mathbb{F}$. Each expression in parentheses on the right-hand sides of these equations is a ‘rational combination’ of $\{a, b, c, d, k\}$, so each also lies in $\mathbb{F}$ by Definition 3. This demonstrates that $\mathbb{F}(\sqrt{k})$ is closed under the rational operations, and is therefore a field. We have now shown that $\mathbb{F}(\sqrt{k})$ is an extension of $\mathbb{F}$ containing $\sqrt{k}$. We must only show that no smaller field has these properties. Suppose $\mathbb{G}$ is a field such that $\mathbb{F} \subset \mathbb{G}$ and $\sqrt{k} \in \mathbb{G}$. For any $a, b \in \mathbb{F}$, we have $a, b, \sqrt{k} \in \mathbb{G}$. Since $\mathbb{G}$ is a field, $a + b\sqrt{k} \in \mathbb{G}$. This implies $\mathbb{F}(\sqrt{k}) \subset \mathbb{G}$. So $\mathbb{F}(\sqrt{k})$ is the *smallest* extension of $\mathbb{F}$ containing k . $\square$

Notice that $\mathbb{F}(\sqrt{k})$ makes sense even if $\sqrt{k} \in \mathbb{F}$. In this case, the defining formula (1) just reduces to $\mathbb{F}$, since $a + b\sqrt{k}$ lies in $\mathbb{F}$ whenever $a, b, \sqrt{k}$ do.

In what follows, we will frequently need to make a statement of the form

$\mathbb{F}$ and $\mathbb{G}$ are real fields, and $\mathbb{G}$ is a quadratic extension of $\mathbb{F}$.

The second clause means that there is some $k \in \mathbb{F}$ for which $\mathbb{G} = \mathbb{F}(\sqrt{k}) \neq \mathbb{F}$. It will be convenient to have a symbolic abbreviation for such a statement. Let us agree to adopt the notation ' $\mathbb{F} \prec \mathbb{G}$ ' for this purpose.

Let us now look at a few examples of quadratic extensions.

1. $\mathbb{Q} \prec \mathbb{Q}(\sqrt{2})$. This obviously fits the definition with $\mathbb{F} = \mathbb{Q}$ and $k = 2$.
2. $\mathbb{Q}(\sqrt{2}) \prec \mathbb{Q}(\sqrt{2})(\sqrt{3})$. Let $\mathbb{F} = \mathbb{Q}(\sqrt{2})$ and $k = 3$. Assume $\sqrt{3} \in \mathbb{F}$ for contradiction. This would mean that $\sqrt{3} = a + b\sqrt{2}$ for some $a, b \in \mathbb{Q}$. Squaring both sides, we get $3 = a^2 + 2ab\sqrt{2} + 2b^2$, or

$$\sqrt{2} = \frac{3 - a^2 - 2b^2}{2ab}.$$

This is a contradiction: the right-hand side is rational, while the left-hand side is irrational. It follows that $\mathbb{F} \prec \mathbb{F}(\sqrt{3})$, as claimed. The elements of the field $\mathbb{F}(\sqrt{3}) = \mathbb{Q}(\sqrt{2})(\sqrt{3})$ are

$$(a + b\sqrt{2}) + (c + d\sqrt{2})\sqrt{3} = a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6},$$

where $a, b, c, d \in \mathbb{Q}$. The right-hand expression here reveals a symmetry between $\sqrt{2}$ and $\sqrt{3}$, and suggests the notation $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2})(\sqrt{3})$, which also equals $\mathbb{Q}(\sqrt{3})(\sqrt{2})$. Note that this is *not* a quadratic extension of $\mathbb{Q}$. It is however a quadratic extension of both $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$.

3. $\mathbb{Q}(\sqrt{2}, \sqrt{3}) \prec \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$. This example works much like the last one, but with $k = 5$. As one readily verifies, the elements of $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ are $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{5} + e\sqrt{6} + f\sqrt{10} + g\sqrt{15} + h\sqrt{30}$ ($a, b, c, d, e, f, g, h \in \mathbb{Q}$).

5. THE CONSTRUCTIBLE NUMBERS. In this section we will make the promised connection between field theory and constructibility. The crucial link between them comes in the following definition, which introduces the notion of a *constructible number*. Such a number is in absolute value the distance between two constructible points. This innocent definition is the key to the whole theory, since it will turn out that the constructible numbers form a field. This field is denoted by ' $\mathbb{E}$ ', in honor of Euclid, who first codified the theory of geometric constructions.

Definition 4. *A real number a is said to be constructible if $|a| = AB$ for some pair of constructible points A and B . The set of all constructible numbers is denoted by $\mathbb{E}$.*

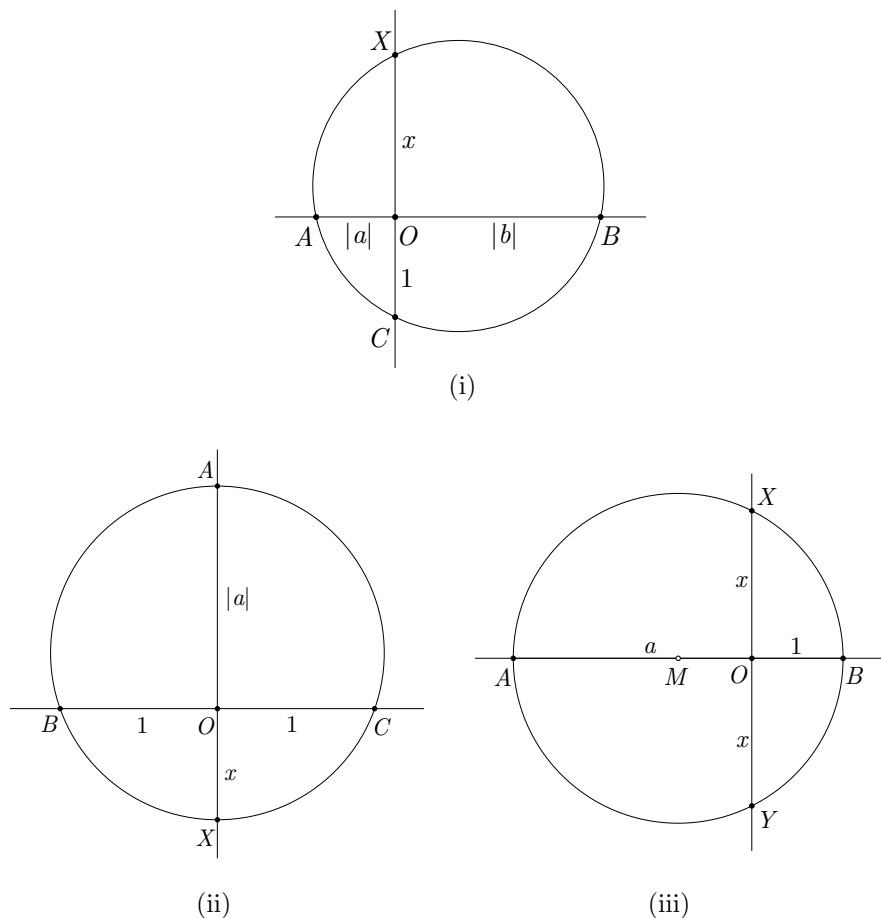
Lemma 1. $a \in \mathbb{E}$ if and only if $(a, 0) \in \mathcal{C}$.

Proof. Suppose $a \in \mathbb{E}$, so that $|a| = AB$ for some $A, B \in \mathcal{C}$. Using Theorem 5, we may construct a length equal to AB on the x -axis, starting from O . There are two directions in which this can be done; one yields a construction of the point $(a, 0)$. Conversely, suppose $(a, 0) \in \mathcal{C}$. Since $(0, 0) \in \mathcal{C}$ as well, clearly $|a|$ is the distance between two constructible points. Hence $a \in \mathbb{E}$ by definition. $\square$

We now prove the critical theorem that $\mathbb{E}$ is a field. In fact, we prove that it is a Euclidean field. The idea of the proof is illustrated in figures 10i-iii.

Theorem 12. *The constructible numbers $\mathbb{E}$ form a Euclidean field.*

Figure 10: Constructions demonstrating that $\mathbb{E}$ is a Euclidean field.



Proof. First, obviously $0 = OO$ and $1 = OI$ belong to $\mathbb{E}$, by Theorem 1. We now prove that $\mathbb{E}$ is closed under the rational operations.

Note first that if $a \in \mathbb{E}$, then $-a \in \mathbb{E}$, since $|-a| = |a|$. Because $\mathbb{E}$ is closed under negation, to prove closure under addition and subtraction we may restrict our attention to the *positive* elements. Thus let $a, b \in \mathbb{E}$ be positive, and without loss of generality suppose $a \geq b$. There must be constructible points A, B, C, D with $a = AB$ and $b = CD$. Use Theorem 6 to copy the distance AB onto $\vec{CD}$, starting from C . The result is a point X such that $CX = a \pm b$. Since $a \pm b$ are distances between pairs of constructible points, $a \pm b \in \mathbb{E}$. This proves closure under addition and subtraction.

Consider multiplication. Let $a, b \in \mathbb{E}$. Certainly $B = (b, 0)$ is constructible, and by an argument exactly like the proof of Lemma 1, $A = (-a, 0)$ is as well. We already know that $C = (0, -1)$ is constructible. Using Theorem 7, construct the circumcircle of $\triangle ABC$, as in figure 10i. This yields the point X shown. By the ‘Crossed Chords Theorem’ of elementary geometry, we have $x = x \cdot 1 = |a| \cdot |b| = |ab|$. Since $|ab|$ is the distance between two constructible points, $ab \in \mathbb{E}$. This proves closure under multiplication.

As for reciprocation, consider figure 10ii. Given a nonzero number $a \in \mathbb{E}$, we construct the points $A = (0, a)$, $B = (-1, 0)$ and $C = (1, 0)$. After drawing the circumcircle of $\triangle ABC$, we have constructed X . Observe that $x \cdot |a| = 1 \cdot 1 = 1$, so that $x = |a|^{-1} = |a^{-1}|$. It follows that $a^{-1} \in \mathbb{E}$, establishing closure under reciprocation (which suffices in place of division).

Finally, let a be a positive element of $\mathbb{E}$. We wish to show that $\sqrt{a} \in \mathbb{E}$, so that $\mathbb{E}$ is a Euclidean field. This is done in figure 10iii. Construct $A = (-a, 0)$ and $B = (1, 0)$, followed by their midpoint M . Next, construct the circle $(M)_A$ and its intersection points with the y -axis, X and Y . By symmetry, note that $OX = OY = x$. Now $x^2 = x \cdot x = a \cdot 1 = a$, so that $x = \sqrt{a}$, and $\sqrt{a} \in \mathbb{E}$. $\square$

We have so far studied two concepts related to fields: quadratic extensions, and the constructible number field. What we would like to do now is make a connection between these ideas. This begins with a second look at a very familiar piece of mathematics — the quadratic formula. Recall that this says

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Evidently, *rational operations and square roots suffice* to compute the solutions.

Consider the problem of solving a quadratic equation ‘in’ a field $\mathbb{F}$. This simply means that the coefficients a, b, c lie in $\mathbb{F}$, and we wish to know whether the solutions also lie in $\mathbb{F}$ (assuming they are real). To get real solutions, we assume $b^2 \geq 4ac$, so that the discriminant $k = b^2 - 4ac$ is nonnegative. Notice that $k \in \mathbb{F}$, since it is a rational combination of a, b, c . The problem is that $\sqrt{k} = \sqrt{b^2 - 4ac}$ may not lie in $\mathbb{F}$. If it does, then the quadratic formula shows that we can solve the equation in $\mathbb{F}$. What if $\sqrt{k} \notin \mathbb{F}$? Then consider $\mathbb{G} = \mathbb{F}(\sqrt{k})$. Since a, b, c belong to $\mathbb{G}$ automatically, as does $\sqrt{k}$, the quadratic formula shows that we can solve the equation in $\mathbb{G}$. To sum up the whole discussion:

Lemma 2. *If a, b, c lie in a field $\mathbb{F}$, with $a \neq 0$ and $b^2 \geq 4ac$, then the solutions of the quadratic equation $ax^2 + bx + c = 0$ are either in $\mathbb{F}$ or at worst in $\mathbb{F}(\sqrt{k})$ where $k = b^2 - 4ac$.*

With this in mind, we make the following definition.

Definition 5. *Let $\mathbb{F}$ be a field.*

1. *An $\mathbb{F}$ -point is a point (a, b) such that $a, b \in \mathbb{F}$.*
2. *An $\mathbb{F}$ -line is a line $ax + by = c$ such that $a, b, c \in \mathbb{F}$.*
3. *An $\mathbb{F}$ -circle is a circle $x^2 + y^2 + ax + by = c$ such that $a, b, c \in \mathbb{F}$.*

We refer to $\mathbb{F}$ -points, $\mathbb{F}$ -lines, and $\mathbb{F}$ -circles collectively as $\mathbb{F}$ -objects. The set of all $\mathbb{F}$ -objects is denoted by $\Omega(\mathbb{F})$.

Now comes a vitally important lemma which relates the set $\Omega(\mathbb{F})$ to the earlier notion of descent, and hence to the theory of constructibility.

Lemma 3 (Descent Lemma). *Let $\mathbb{F}$ be a field.*

1. *If A and B are distinct points in $\Omega(\mathbb{F})$, their descendants are in $\Omega(\mathbb{F})$.*
2. *If ℓ_1 and ℓ_2 are distinct intersecting lines in $\Omega(\mathbb{F})$, their unique descendant is in $\Omega(\mathbb{F})$.*
3. *If ℓ and c are an intersecting line and circle in $\Omega(\mathbb{F})$, their descendants are either in $\Omega(\mathbb{F})$ or at worst in $\Omega(\mathbb{G})$, where $\mathbb{F} \prec \mathbb{G}$.*
4. *If c_1 and c_2 are distinct intersecting circles in $\Omega(\mathbb{F})$, their descendants are either in $\Omega(\mathbb{F})$ or at worst in $\Omega(\mathbb{G})$, where $\mathbb{F} \prec \mathbb{G}$.*

Proof. We give a brief outline of the proof for each statement. The fact that fields are closed under rational combinations will be used without comment.

1. Let $a, b, c, d \in \mathbb{F}$, so $(a, b), (c, d) \in \Omega(\mathbb{F})$. The line through (a, b) and (c, d) is $(d - b)x + (a - c)y = ad - bc$, an $\mathbb{F}$ -line. The circle with center (a, b) containing (c, d) is $x^2 + y^2 - 2ax - 2by = c^2 + d^2 - 2(ac + bd)$, an $\mathbb{F}$ -circle.
2. The intersection point of the $\mathbb{F}$ -lines $ax + by = p$ and $cx + dy = q$ is

$$\left(\frac{pd - bq}{ad - bc}, \frac{aq - pc}{ad - bc} \right) \in \Omega(\mathbb{F}).$$

3. Consider an $\mathbb{F}$ -line $ax + by = p$ and an $\mathbb{F}$ -circle $x^2 + y^2 + cx + dy = q$, which we assume intersect. Either a or b must be nonzero, so the equation of the line may be solved for one of the variables, say y . This expression for y may be substituted in the equation of the circle, leading to a quadratic equation in x . Let k be the discriminant of that equation. If $\sqrt{k} \in \mathbb{F}$, the solutions for x lie in $\mathbb{F}$. The expression $y = b^{-1}(p - ax)$ shows that the solutions for y also lie in $\mathbb{F}$. Hence the intersection points (x, y) lie in $\Omega(\mathbb{F})$. If $\sqrt{k} \notin \mathbb{F}$, then let $\mathbb{G} = \mathbb{F}(\sqrt{k})$. By Lemma 2, the solutions for x , and hence for y , lie in $\mathbb{G}$. So the intersection points (x, y) are in $\Omega(\mathbb{G})$.

4. Consider $\mathbb{F}$ -circles $x^2 + y^2 + ax + bx = p$ and $x^2 + y^2 + cx + dy = q$, which we assume intersect. If they are tangent, their intersection point lies on the $\mathbb{F}$ -line through their centers, $2(b - d)x + 2(c - a)y = ad - bc$. Replacing one of the circles with this line brings us back to case 3. If the circles have two intersection points (x_1, y_1) and (x_2, y_2) , these points must satisfy the equation $(c - a)x + (d - b)y = q - p$, obtained by subtracting the equations of the two circles. This is an $\mathbb{F}$ -line through (x_1, y_1) and (x_2, y_2) . Replacing one circle with this line, we are back to case 3. $\square$

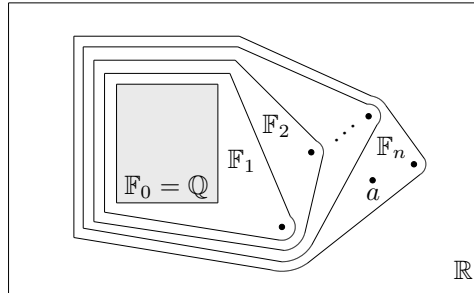
We are finally ready to present the main result of this section, known as the *Constructibility Theorem*. This gives a necessary condition in terms of quadratic extensions for a number a to be constructible. Informally, it says that if a number a is constructible, we must be able to ‘capture’ a by starting with the rational field $\mathbb{Q}$ and making a finite number of successive quadratic extensions.

Theorem 13 (Constructibility Theorem). *If $a \in \mathbb{R}$ is constructible, then there is an increasing sequence of quadratic extensions*

$$\mathbb{Q} = \mathbb{F}_0 \prec \mathbb{F}_1 \prec \mathbb{F}_2 \prec \cdots \prec \mathbb{F}_n$$

starting with the rationals, such that $a \in \mathbb{F}_n$ (see figure 11).

Figure 11: Schematic diagram of the Constructibility Theorem.



Proof. Suppose a is constructible. By Lemma 1, $(a, 0)$ is a constructible point. Hence there is a valid sequence $\langle \omega_0, \omega_1, \omega_2, \dots, \omega_N \rangle$, where $\omega_0 = O$, $\omega_1 = I$, and $\omega_N = (a, 0)$. Notice that $\omega_0 = O \in \Omega(\mathbb{Q}) = \Omega(\mathbb{F}_0)$. It is possible that all the ω_i belong to $\Omega(\mathbb{F}_0)$, and $\omega_N = (a, 0)$ in particular. In this case $a \in \mathbb{F}_0$, and the conclusion of the theorem is trivially true with $n = 0$. Otherwise, there must be a first index i such that $\omega_i \notin \Omega(\mathbb{F}_0)$. Since ω_i is descended from the objects preceding it in the sequence, all of which lie in $\Omega(\mathbb{F}_0)$, the Descent Lemma implies that there must be a quadratic extension $\mathbb{F}_1$ of $\mathbb{F}_0$ such that $\omega_i \in \Omega(\mathbb{F}_1)$. Now either $\omega_N \in \Omega(\mathbb{F}_1)$ (so that $a \in \mathbb{F}_1$) or there must be a first index $j > i$ for which $\omega_j \notin \Omega(\mathbb{F}_1)$. The Descent Lemma then supplies a quadratic extension $\mathbb{F}_2$ of $\mathbb{F}_1$ such that $\omega_j \in \Omega(\mathbb{F}_2)$. Continuing this way, we soon reach ω_N . This process produces a series of quadratic extensions $\mathbb{Q} = \mathbb{F}_0 \prec \mathbb{F}_1 \prec \mathbb{F}_2 \prec \cdots \prec \mathbb{F}_n$ such that $\omega_N \in \Omega(\mathbb{F}_n)$, and hence $a \in \mathbb{F}_n$. $\square$

6. DOUBLING THE CUBE. We are finally ready to apply our algebraic theorems to the first of the ancient construction problems: doubling the cube. This problem requires a construction for one edge of a cube having twice the volume of a given cube. For simplicity, let us scale our units so that the original cube has edges of length 1. Then to produce a cube of volume 2, we must construct an edge of length $\sqrt[3]{2}$. The impossibility of such a construction will be established by showing that $\sqrt[3]{2}$ does not belong to the field $\mathbb{E}$ of constructible numbers. We will need two lemmas to reach this result. The first is a method for finding the rational zeroes of a polynomial with integer coefficients. [A number a is said to be a *zero* of the function $f(x)$ if we have $f(a) = 0$.]

Lemma 4 (Rational Zeroes Lemma). *Let $f(x) = a_0x^d + a_1x^{d-1} + \cdots + a_d$ be a nonzero polynomial with integer coefficients. If p and q are relatively prime integers such that $f(p/q) = 0$, then p must divide a_d and q must divide a_0 .*

Proof. Since p and q are relatively prime, they have no common prime factors, and the rational number p/q is in ‘lowest terms’. Also, p^d and q are relatively prime, as are p and q^d . The reason is that p^d cannot have any prime factors that were not already present in p . With this in mind, consider the equation $f(p/q) = 0$. After multiplying through by q^d to clear denominators, this says

$$a_0p^d + a_1p^{d-1}q + a_2p^{d-2}q^2 + \cdots + a_{d-1}pq^{d-1} + a_dq^d = 0.$$

Isolating the last term and factoring out a p , we find that

$$a_dq^d = -p(a_0p^{d-1} + a_1p^{d-2}q + \cdots + a_{d-1}q^{d-1}).$$

Since p is explicitly a factor of the right-hand side, it must also divide the left-hand side. But p has no prime factors in common with q^d . Hence the only possibility is that p divides a_d . A similar argument, with the first term isolated instead of the last, shows that q must divide a_0 . $\square$

The Rational Zeroes Lemma has two main uses. The more practical one is to make a finite list of possible candidates for the rational roots of a given polynomial. The list is finite because a_0 and a_d have only finitely many divisors. Each candidate may then be substituted directly into the polynomial, and any zeroes that may exist will eventually be discovered in the process. For example, the candidates for rational zeroes of $f(x) = 6x^3 + x^2 - 11x - 6$ are

$$\pm 6, \pm 3, \pm 2, \pm \frac{3}{2}, \pm 1, \pm \frac{2}{3}, \pm \frac{1}{2}, \pm \frac{1}{3}, \pm \frac{1}{6}.$$

By direct substitution, it can be shown that the only zeroes of $f(x)$ in this list are $\frac{3}{2}$, -1 , and $-\frac{2}{3}$. The Rational Zeroes Lemma then assures us that there can be no other rational zeroes. A second theoretical use of the lemma is in proving that a given number is irrational. Suppose a is a real number, and that we know a polynomial $f(x)$ with integer coefficients such that $f(a) = 0$. If we try the procedure just explained, and it turns out that $f(x)$ has no rational zeroes, then it follows that a must be irrational.

We may use this idea to prove that $\sqrt{2}$ and $\sqrt[3]{2}$ are irrational. Thus, $\sqrt{2}$ is a zero of $f(x) = x^2 - 2$, which has integer coefficients. According to the Rational Zeroes Lemma, the only candidates for rational zeroes are ± 2 and ± 1 . But none of these possibilities actually works. Hence $\sqrt{2}$ is irrational. An exactly similar argument using $f(x) = x^3 - 2$ shows that $\sqrt[3]{2}$ is irrational.

Our final lemma says in essence that $\sqrt[3]{2}$ is not able to be ‘captured’ by a quadratic extension unless it was already present in the original field.

Lemma 5 (Capture Lemma I). *If $\mathbb{F} \prec \mathbb{G}$ and $\sqrt[3]{2} \in \mathbb{G}$, then in fact $\sqrt[3]{2} \in \mathbb{F}$.*

Proof. Let k be a nonnegative number such that $\mathbb{G} = \mathbb{F}(\sqrt{k})$. [Recall that $k \in \mathbb{F}$, but $\sqrt{k} \notin \mathbb{F}$.] Then by hypothesis we have $\sqrt[3]{2} = a + b\sqrt{k}$ for some $a, b \in \mathbb{F}$. If $b = 0$, then $\sqrt[3]{2} = a \in \mathbb{F}$, and we have reached the desired conclusion. Otherwise, $b \neq 0$. By cubing both sides of our expression for $\sqrt[3]{2}$, we find that

$$2 = a^3 + 3a^2b\sqrt{k} + 3ab^2k + b^3k\sqrt{k} = (a^3 + 3ab^2k) + (3a^2b + b^3k)\sqrt{k}.$$

Consider the second parenthesized expression on the right-hand side. If this was nonzero, then we would be able to isolate $\sqrt{k}$, getting

$$\sqrt{k} = \frac{2 - (a^3 + 3ab^2k)}{3a^2b + b^3k}.$$

This says that $\sqrt{k}$ is a rational combination of numbers in $\mathbb{F}$, and is therefore itself in $\mathbb{F}$ — a contradiction. So in fact we must have

$$3a^2b + b^3k^2 = 0 \quad \text{and} \quad a^3 + 3ab^2k^2 = 2.$$

Since $b \neq 0$, we can divide the first equation by b . Thus $3a^2 + b^2k^2 = 0$, or $b^2k^2 = -3a^2$. Substituting this in the second equation gives $a^3 + 3a(-3a^3) = 2$, or $-8a^3 = 2$. Write $c = -2a$. Then $c^3 = -8a^3 = 2$, and we must have $c = \sqrt[3]{2}$. Note that $c = -2a \in \mathbb{F}$, since $a \in \mathbb{F}$. Hence $\sqrt[3]{2} \in \mathbb{F}$, as desired. $\square$

We are finally ready to prove the impossibility of doubling the cube!

Theorem 14 (Doubling the Cube). *It is impossible to double the cube with a straightedge and a compass. More formally, $\sqrt[3]{2}$ is not a constructible number.*

Proof. Assume for contradiction that $\sqrt[3]{2} \in \mathbb{E}$. Applying the Constructibility Theorem, there must be an increasing sequence of quadratic extensions

$$\mathbb{Q} = \mathbb{F}_0 \prec \mathbb{F}_1 \prec \mathbb{F}_2 \prec \cdots \prec \mathbb{F}_n$$

such that $\sqrt[3]{2} \in \mathbb{F}_n$. In the Capture Lemma I above, take $\mathbb{G} = \mathbb{F}_n$ and $\mathbb{F} = \mathbb{F}_{n-1}$. This is consistent with the hypotheses of the lemma, since $\mathbb{F}_{n-1} \prec \mathbb{F}_n$ and $\sqrt[3]{2} \in \mathbb{F}_n$. We conclude that $\sqrt[3]{2}$ must belong to $\mathbb{F}_{n-1}$. This however allows us to apply Lemma 5 again, this time with $\mathbb{G} = \mathbb{F}_{n-1}$ and $\mathbb{F} = \mathbb{F}_{n-2}$. Here again, the conclusion is that $\sqrt[3]{2} \in \mathbb{F}_{n-2}$. Continuing this process, we eventually see that $\sqrt[3]{2} \in \mathbb{F}_0 = \mathbb{Q}$, a result we know to be false by the Rational Zeroes Lemma. The contradiction rules out the doubling of the cube. $\square$

7. TRISECTING THE ANGLE. Let us now consider the trisection of the angle. For the sake of convenience, we will extend our terminology and speak of ‘constructible angles’ and ‘constructible rays’. Up to now, only points, lines and circles could be constructible. By a *constructible angle*, we mean an ordered triple of noncollinear constructible points, ordered with the vertex in the middle. By a *constructible ray*, we mean an ordered pair of constructible points, ordered with the endpoint first. Given a constructible angle, we seek a construction for the two rays inside the angle which divide it into three equal parts. Notice that the construction sought here is a relative one, since we must assume a known construction for the original angle. [If the angle itself cannot be constructed, it seems pointless to ask for its trisectors to be constructed.] A good angle for our purposes is 60° . Such an angle is constructible as follows: draw the circles $(O)_I$ and $(I)_O$ and mark one of the intersection points, say X . Then $\triangle OIX$ is equilateral, and $\angle XOI = 60^\circ$. If every constructible angle could be trisected with straightedge and compass, one would then have a construction for some 20° angle. Our goal is to prove that in fact no 20° angle is constructible.

Lemma 6 (Triple Angle Formula). $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.

Proof. Using the addition formulas for sine and cosine, and the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$, we may expand $\cos 3\theta = \cos(2\theta + \theta)$ as follows:

$$\begin{aligned}\cos 3\theta &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (\cos^2\theta - \sin^2\theta) \cos \theta - (2\sin\theta \cos\theta) \sin \theta \\ &= (2\cos^2\theta - 1) \cos \theta - 2(1 - \cos^2\theta) \cos \theta \\ &= 4\cos^3\theta - 3\cos\theta. \quad \square\end{aligned}$$

Lemma 7. *The number $\cos 20^\circ$ is a zero of the cubic $f(x) = 8x^3 - 6x - 1$.*

Proof. Substitute $\theta = 20^\circ$ in the Triple Angle Formula (Lemma 6) to get

$$\cos 60^\circ = 4(\cos 20^\circ)^3 - 3(\cos 20^\circ).$$

Noting that $\cos 60^\circ = \frac{1}{2}$, we multiply through by 2 and rearrange terms to get

$$8(\cos 20^\circ)^3 - 6(\cos 20^\circ) - 1 = 0.$$

Hence $\cos 20^\circ$ is a zero of the cubic $f(x) = 8x^3 - 6x - 1$, as desired. $\square$

An important fact about the cubic f introduced in Lemma 7 is that *it has no rational zeroes*. The reason for this is that the Rational Zeroes Lemma limits the possible candidates to ± 1 , $\pm \frac{1}{2}$, $\pm \frac{1}{4}$, and $\pm \frac{1}{8}$. By simple computations, one shows directly that none of the candidates actually work.

The last pre-requisite needed to rule out the trisection of the angle is a second ‘capture lemma’, like Lemma 5 of the previous section. Its purpose is to ensure that a zero of f cannot be captured by a quadratic extension unless some zero of f was already present in the original field.

Lemma 8 (Capture Lemma II). *If $\mathbb{F} \prec \mathbb{G}$ and f has a zero in $\mathbb{G}$, then f also has a zero in $\mathbb{F}$.*

Proof. Let k be a nonnegative number such that $\mathbb{G} = \mathbb{F}(\sqrt{k})$. [Recall that $k \in \mathbb{F}$, but $\sqrt{k} \notin \mathbb{F}$.] Then by hypothesis, some number of the form $a + b\sqrt{k}$ with $a, b \in \mathbb{F}$ satisfies the equation $f(a + b\sqrt{k}) = 0$. If $b = 0$, then this zero of f is simply a , which lies in $\mathbb{F}$ and satisfies the conclusion of the lemma. Thus assume $b \neq 0$, and write the hypothesis in the form

$$8(a + b\sqrt{k})^3 - 6(a + b\sqrt{k}) - 1 = 0.$$

Multiplying this out and separating off the terms containing $\sqrt{k}$, we have

$$(8a^3 + 24ab^2k - 6a - 1) + (24a^2b + 8b^3k - 6b)\sqrt{k} = 0.$$

If the second of the parenthesized expressions here was nonzero, then we would be able to express $\sqrt{k}$ as a rational combination of $a, b, k \in \mathbb{F}$. That would be a contradiction, since it would imply $\sqrt{k} \in \mathbb{F}$. The second parenthesized expression must accordingly equal 0, and so therefore must the first. In short,

$$8a^3 + 24ab^2k - 6a - 1 = 0 \quad \text{and} \quad 24a^2b + 8b^3k - 6b = 0. \quad (2)$$

Dividing the second equation in (2) by $2b$ and rearranging the terms, we get $4b^2k = 3 - 12a^2$. This in turn may be multiplied through by $6a$ to get

$$24ab^2k = 6a(3 - 12a^2) = 18a - 72a^3. \quad (3)$$

Substitute the expression of (3) into the first equation in (2). This gives

$$8a^3 + (18a - 72a^3) - 6a - 1 = 0, \quad \text{or} \quad -64a^3 + 12a - 1 = 0.$$

This last result can be written as $8(-2a)^3 - 6(-2a) - 1 = 0$. Let $c = -2a$. Then $f(c) = 8c^3 - 6c - 1 = 0$, so that c is a zero of f . Moreover, $c = -2a \in \mathbb{F}$. $\square$

Now on to the impossibility proof!

Theorem 15 (Trisecting the Angle). *It is impossible to trisect the angle with a straightedge and a compass. More formally, no 20° angle is constructible.*

Proof. Suppose we could construct an angle $\angle B$ such that $\angle B = 20^\circ$. Using the construction techniques of Theorems 3 and 5, we could then construct a right triangle $\triangle ABC$ with $AB = 1$ and $\angle C = 90^\circ$, as well as $\angle B = 20^\circ$. Clearly $BC = \cos 20^\circ$, so that $\cos 20^\circ$ is the distance between two constructible points and lies in $\mathbb{E}$. Applying the Constructibility Theorem, we get an increasing sequence of quadratic extensions $\mathbb{Q} = \mathbb{F}_0 \prec \mathbb{F}_1 \prec \mathbb{F}_2 \prec \cdots \prec \mathbb{F}_n$ such that $\cos 20^\circ \in \mathbb{F}_n$. By Lemma 7, this means f has a zero in $\mathbb{F}_n$. In the Capture Lemma II, take $\mathbb{G} = \mathbb{F}_n$ and $\mathbb{F} = \mathbb{F}_{n-1}$. This is consistent with the hypotheses of the lemma, since $\mathbb{F}_{n-1} \prec \mathbb{F}_n$ and $\mathbb{F}_n$ contains a zero of f . We conclude that $\mathbb{F}_{n-1}$ must contain a zero of f . This however allows us to apply Lemma 8 again, this time with $\mathbb{G} = \mathbb{F}_{n-1}$ and $\mathbb{F} = \mathbb{F}_{n-2}$. One concludes that $\mathbb{F}_{n-2}$ contains a zero of f . Continuing this process, we eventually see that $\mathbb{F}_0 = \mathbb{Q}$ contains a zero of f . Since f has no rational zeroes, we have reached a contradiction. $\square$

8. ALGEBRAIC NUMBERS. In order to analyze the problem of squaring the circle, we must digress awhile to consider the subject of algebraic numbers. This subject is a vast and very deep part of modern algebra, but for our purposes a brief introduction will suffice. We begin by defining what it means for a complex number a to be algebraic ‘over’ a field $\mathbb{F}$.

Definition 6. Let $\mathbb{F}$ be a field. A number $a \in \mathbb{C}$ is said to be algebraic over $\mathbb{F}$ if there is at least one polynomial

$$f(x) = c_0x^d + c_1x^{d-1} + \cdots + c_{d-1}x + c_d,$$

with coefficients $c_0, c_1, \dots, c_d \in \mathbb{F}$, such that $f(a) = 0$. The set of all complex numbers that are algebraic over $\mathbb{F}$ is denoted by $\mathbb{A}(\mathbb{F})$.

A complex number a is called simply ‘algebraic’ if it is algebraic over $\mathbb{Q}$. The set $\mathbb{A}(\mathbb{Q})$ is usually denoted by the abbreviated symbol $\mathbb{A}$. It is worth noting that if $a \in \mathbb{A}$, then the coefficients $c_0, c_1, \dots, c_d$ appearing in Definition 6 can be restricted to integer values without loss of generality. The reason is that if $c_0, c_1, \dots, c_d$ are rational fractions in ‘lowest terms’, we may multiply the equation $c_0a^d + c_1a^{d-1} + \cdots + c_d = 0$ by the least common multiple of all the denominators to get a new equation $C_0a^d + C_1a^{d-1} + \cdots + C_d = 0$, in which the coefficients are now integers.

It turns out that the set $\mathbb{A}(\mathbb{F})$ is actually a field. This result, while interesting, is quite difficult to prove. Since it will not be needed in this article, we will not study it further. It is important to bear in mind that the notation ‘ $\mathbb{A}(\mathbb{F})$ ’ is being used here simply for the *set* of numbers algebraic over $\mathbb{F}$, with no presumption of this set being a *field*. A very simple lemma about $\mathbb{A}(\mathbb{F})$ is as follows.

Lemma 9. For any field $\mathbb{F}$, we have $\mathbb{F} \subset \mathbb{A}(\mathbb{F})$.

Proof. Take $a \in \mathbb{F}$. Note that a is a zero of the polynomial $f(x) = x - a$, whose coefficients 1 and $-a$ lie in $\mathbb{F}$. By definition, this means that $a \in \mathbb{A}(\mathbb{F})$. $\square$

In what follows, we are concerned only with *real* algebraic numbers, such as

$$-\frac{7}{3}, \quad \sqrt{2}, \quad \sqrt[5]{2/9}, \quad \sqrt{2\sqrt{2} + \sqrt[3]{3}}.$$

To show that any of these numbers is algebraic, we must find a polynomial with integer coefficients having that number as a zero. For the first three examples, this is simple: $3x + 7$, $x^2 - 2$, and $9x^5 - 2$. The last example is a little trickier. Here is one way to derive the answer:

$$\begin{aligned} x = \sqrt{2\sqrt{2} + \sqrt[3]{3}} &\Rightarrow x^2 = 2\sqrt{2} + \sqrt[3]{3} \\ &\Rightarrow (x^2 - 2\sqrt{2})^3 = 3 \\ &\Rightarrow x^6 - 6\sqrt{2}x^4 + 24x^2 - 16\sqrt{2} = 3 \\ &\Rightarrow x^6 + 24x^2 - 3 = \sqrt{2}(6x^4 + 16) \\ &\Rightarrow (x^6 + 24x^2 - 3)^2 = 2(6x^4 + 16)^2 \\ &\Rightarrow x^{12} - 24x^8 - 6x^6 + 192x^4 - 144x^2 - 503 = 0. \end{aligned}$$

The following theorem shows why the algebraic numbers bear relevance to the subject of constructibility.

Theorem 16 (Constructible Implies Algebraic). *Every constructible number is algebraic. That is, $\mathbb{E} \subset \mathbb{A}$.*

Proof. Take $a \in \mathbb{E}$. By the Constructibility Theorem, there is an increasing sequence of quadratic extensions $\mathbb{Q} = \mathbb{F}_0 \prec \mathbb{F}_1 \prec \mathbb{F}_2 \prec \cdots \prec \mathbb{F}_n$ such that $a \in \mathbb{F}_n$. Since $\mathbb{F}_n \subset \mathbb{A}(\mathbb{F}_n)$ by Lemma 9, we have $a \in \mathbb{A}(\mathbb{F}_n)$. This fact will be used as the base case for a ‘backward induction’, in which we show that $a \in \mathbb{A}(\mathbb{F}_{i+1})$ implies $a \in \mathbb{A}(\mathbb{F}_i)$ [notice how the index *decreases* by one, whereas in ordinary induction it would increase by one]. Thus assume for induction that $a \in \mathbb{A}(\mathbb{F}_{i+1})$. This means that there is a polynomial

$$f(x) = c_0x^d + c_1x^{d-1} + \cdots + c_d$$

with coefficients in $\mathbb{F}_{i+1}$, such that $f(a) = 0$. Now since $\mathbb{F}_i \prec \mathbb{F}_{i+1}$, for some nonnegative $k \in \mathbb{F}_i$ we can write $\mathbb{F}_{i+1} = \mathbb{F}_i(\sqrt{k})$. It follows that

$$c_j = a_j + b_j\sqrt{k} \quad j = 0, 1, \dots, d,$$

where all the numbers a_j and b_j lie in $\mathbb{F}_i$. The equation $f(a) = 0$ is therefore

$$(a_0 + b_0\sqrt{k})a^d + (a_1 + b_1\sqrt{k})a^{d-1} + \cdots + (a_d + b_d\sqrt{k}) = 0.$$

This we rearrange as follows:

$$a_0a^d + a_1a^{d-1} + \cdots + a_d = -\sqrt{k}(b_0a^d + b_1a^{d-1} + \cdots + b_d).$$

Squaring both sides, we get

$$(a_0a^d + a_1a^{d-1} + \cdots + a_d)^2 = k(b_0a^d + b_1a^{d-1} + \cdots + b_d)^2.$$

This shows that a is a zero of the polynomial

$$g(x) = (a_0x^d + a_1x^{d-1} + \cdots + a_d)^2 - k(b_0x^d + b_1x^{d-1} + \cdots + b_d)^2.$$

When g is multiplied out and simplified, its coefficients will have resulted from the numbers $\{a_0, \dots, a_d, b_0, \dots, b_d, k\}$ by just additions and multiplications. Since all these numbers lie in the field $\mathbb{F}_i$, so too must the coefficients of g . In short, a is a zero of some polynomial with coefficients in $\mathbb{F}_i$. Hence $a \in \mathbb{A}(\mathbb{F}_i)$, completing the backward induction. The induction soon takes us from the base case $a \in \mathbb{A}(\mathbb{F}_n)$ to the required conclusion $a \in \mathbb{A}(\mathbb{F}_0) = \mathbb{A}(\mathbb{Q}) = \mathbb{A}$. $\square$

9. SQUARING THE CIRCLE. The problem of squaring the circle turned out to be much more difficult to resolve than the other two ancient construction problems. The reason for this is that the circular area formula involves the number π , which is a much more complicated number than $\sqrt[3]{2}$ or $\cos 20^\circ$. The difference is that the latter two numbers are algebraic (as witness $x^3 - 2$ and $8x^3 - 6x - 1$), while π is *not* algebraic. This is the famous theorem of Lindemann to which we alluded in the introduction. Numbers that are not algebraic are called *transcendental*. So Lindemann’s theorem is this:

Theorem 17 (Lindemann’s Theorem). *π is a transcendental number.*

The proof of Lindemann’s Theorem is much too difficult to include in this article. Since it is not properly concerned with constructibility anyway, let us allow ourselves to simply take this theorem for granted. It will then be relatively simple to deduce the impossibility of squaring the circle.

Recall that the problem requires a construction for the vertices of a square whose area is the same as that of a given circle. We may as well scale our units so that the given circle has radius 1. Then the area of the circle is equal to π . Finally, here is the result we have been after.

Theorem 18 (Squaring the Circle). *It is impossible to square the circle with a straightedge and a compass. More formally, one cannot construct the vertices of a square of area π .*

Proof. Assume for contradiction that we can construct the vertices of a square of area π . One edge of this square has length $\sqrt{\pi}$, which is therefore a constructible number. That is, $\sqrt{\pi} \in \mathbb{E}$. Since $\mathbb{E}$ is a field (Theorem 12), we also have

$$\pi = \sqrt{\pi} \cdot \sqrt{\pi} \in \mathbb{E}.$$

Applying Theorem 16 (‘Constructible Implies Algebraic’), this means that π is algebraic. Of course, Lindemann’s Theorem says exactly that π is not algebraic! The contradiction shows that squaring the circle is impossible. $\square$

10. CONCLUSION. This concludes our study of constructibility theory as it pertains to the three famous construction problems of antiquity. However, the remarkable resolution of these three problems is far from the whole story. Many other interesting construction problems have been settled by the methods we have studied here. A notable one was to determine all integers $n \geq 3$ for which the vertices of a regular n -sided polygon are constructible. The solution, due half to Gauss and half to Wantzel, turned out to involve ‘Fermat primes’, i.e., primes of the form $2^{2^r} + 1$. When $r = 2$, the number $2^{2^2} + 1 = 17$ is prime, and Gauss not only proved the possibility of constructing the vertices of a regular 17-sided polygon, but actually worked out a specific construction method (which is too lengthy to carry out in practice). He was so proud of this achievement that he asked a stonemason to carve a regular 17-gon on his tombstone. The mason declined, reasoning that the result would look just like a circle! [This is an example of the practical wisdom that mathematicians sometimes lack.] Although this general subject is no longer actively researched, its classical theorems remain to this day among the most charming in mathematics.

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PROBLEMS

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20081. Convex pentagon $ABCDE$ has $AB = BC = 17$, $CD = DE = 15$, and $EA = 18$. If $\cos(\angle EAB) = \frac{13}{85}$ and $\cos(\angle ABC) = -\frac{161}{289}$, compute $[ABCDE]$.

20082. Describe (with proof) all polynomials f for which the following is an identity: $f(x+1) = 1 + f(x) + f'(x)$. Here, f' is the derivative of f .

20083. Let P be the set of all pythagorean triples of the form $(a, a+1, a+k)$, where a and k are positive integers. If the elements of P are ordered according to the value of $a+k$, in ascending order, compute the first four values of k .

20084. $\triangle ABC$ has a right angle at C , legs of lengths a and b , and a hypotenuse of length c , where a, b, c are positive integers. The segment BA is extended through A to a point D so that $AD = 1/c$. A point E is drawn so that $\angle ADE$ is right, $AE = a \cdot b$, and $BE = d$, where d is a positive integer. Prove that $d+1$ and $d-1$ cannot be a pair of 'twin primes'.

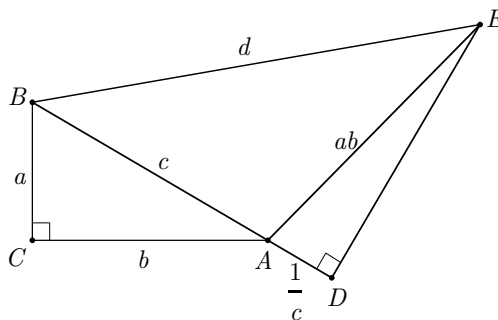


Figure 1: Problem 20084.

20085. Prove that for all monic cubic polynomial with integer coefficients, there exist infinitely many lines that intersect its graph at three distinct lattice points. Recall that a polynomial is called *monic* if its leading coefficient is 1, and a *lattice point* is a point both of whose coordinates are integers.

20086. Compute the number of real solutions (p, q) of the following system:

$$\begin{cases} p^2 = 25 - q^2 \\ p^2 + 2p = q^4 + 8q^2 + 15 \end{cases}$$

20087. Determine (with proof) the largest integer k such that $p^4 \equiv 1 \pmod{k}$ for all primes $p > 5$.

20082. Let $g(x) = f(x)$. The equation is now

$$\int_0^{x+1} g(t) dt - 1 = \int_0^x g(t) dt + g(x).$$
$$\int_x^{x+1} g(t) dt = g(x) + 1.$$

Referring to the diagram below, we may interpret this equation to mean that the area under $y = g(x)$ from x to $x + 1$ is 1 more than the area of the rectangle with base 1 and height $g(x)$. Hence the region above this rectangle must have area 1 for all x . The only such polynomials g are straight lines of slope 2, that is, $g(x) = 2x + b$. Antidifferentiation now gives the answer, $f(x) = x^2 + bx + c$.

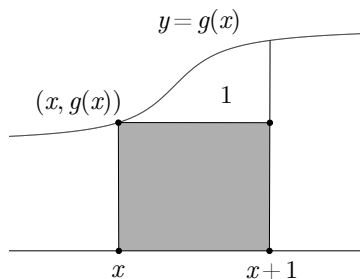


Figure 2: Solution of 20082.

20083. Consider the well-known formulas for pythagorean triples:

$$(m^2 - n^2, 2mn, m^2 + n^2) \quad \text{and} \quad (2mn, m^2 - n^2, m^2 + n^2).$$

In order to make the legs differ by 1, we must have

$$2mn - (m^2 - n^2) = \pm 1, \quad \text{or} \quad (n + m)^2 - 2m^2 = \pm 1.$$

The last formula is a Pell equation. When the sign is $+$, the solutions are generated by $(3 + 2\sqrt{2})^n$. Taking $n = 1$ gives $(3, 4, 5)$. Taking $n = 2$ gives $m = 12$ and $n = 5$, leading to the triple $(119, 120, 169)$. When the sign is $-$, the solutions are generated by $(1 + \sqrt{2})(3 + 2\sqrt{2})^n$. Taking $n = 1$ gives $(20, 21, 29)$. Taking $n = 2$ gives $m = 29$ and $n = 12$, leading to the triple $(696, 697, 985)$. Thus the first four values of k are **2, 9, 50, 289**.

20084. Let $\theta = \angle BAE$. Then:

$$\cos(\pi - \theta) = \frac{1}{abc}, \quad \text{or} \quad \cos \theta = -\frac{1}{abc}.$$

By the law of cosines on $\triangle BAE$, we have $d^2 = c^2 + (ab)^2 - 2(ab)c \cos \theta$. On substituting our expression for $\cos \theta$, as well as $c^2 = a^2 + b^2$, we get:

$$\begin{aligned} a^2 + b^2 + a^2b^2 &= d^2 + 2abc \left(-\frac{1}{abc} \right), \\ a^2 + b^2 + a^2b^2 + 1 &= d^2 - 1, \\ (a^2 + 1)(b^2 + 1) &= (d + 1)(d - 1). \end{aligned}$$

If $d + 1$ and $d - 1$ are both prime, then each is one more than a perfect square. This is a contradiction, however, since no two perfect squares differ by 2. $\square$

20085. Fix a monic cubic polynomial, $y = x^3 + ax^2 + bx + c$. We want to show that infinitely many lines intersect its graph G at three distinct lattice points. Consider the line ℓ with equation $y = mx + k$, where m and k are as yet unspecified integers. Intersection points of G and ℓ satisfy both equations, and are therefore roots of the cubic equation $x^3 + ax^2 + (b - m)x + (c - k) = 0$, obtained by subtracting the equations of G and ℓ . Let p, q, r be three integers such that $p + q + r = -a$ and $p < q < r$, but otherwise arbitrary. If these are to be roots of the required cubic equation, we must have $b - m = pq + qr + rp$ and $c - k = -pqr$. Thus let us simply define m and k in the first place by $m = b - pq - qr - rp$ and $k = c + pqr$. Then $(p, mp + k)$, $(q, mq + k)$, and $(r, mr + k)$ are distinct lattice points on both G and ℓ . Obviously there are infinitely many choices for p, q, r that meet the stated conditions. $\square$

20086. Let us graph the equations in the pq -plane. First, $p^2 = 25 - q^2$ is simply the circle $p^2 + q^2 = 25$, which has center $(0, 0)$ and radius 5. Next observe that $p^2 + 2p = q^4 + 8q^2 + 15$ may be simplified as follows:

$$\begin{aligned} p^2 + 2p + 1 &= q^4 + 8q^2 + 16 \\ (p + 1)^2 &= (q^2 + 4)^2 \\ p + 1 &= \pm(q^2 + 4) \end{aligned}$$

This is the union of two parabolas, $p = q^2 + 3$ and $p = -q^2 - 5$. Graphing the curves as below, it is clear that the number of intersection points is **3**.

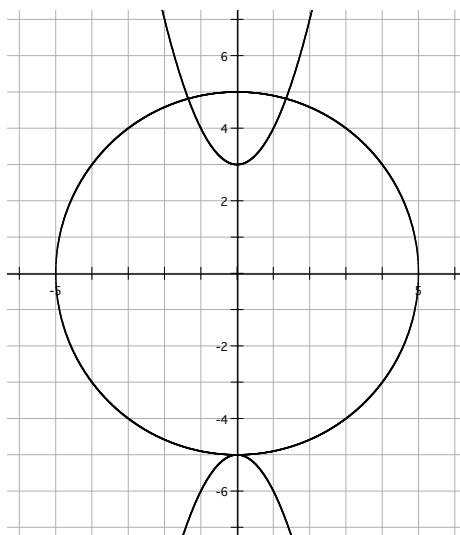


Figure 3: Solution of 20086.

20087. By definition, $p^4 \equiv 1 \pmod{k}$ means that k divides $p^4 - 1$. That is, k divides

$$(p+1)(p-1)(p^2+1).$$

We will first consider divisibility by 2, 3, 5, the primes *not* > 5 . Observe that $p+1$, $p-1$, and p^2+1 are all even, and either $p+1$ or $p-1$ is a multiple of 4 (since they are even number that differ by 2). This accounts for a factor of $16 = 2^4$ in the product. Since p is not a multiple of 3, by the ‘pigeonhole principle’, either $p+1$ or $p-1$ must be a multiple of 3. Also, all primes $p > 5$ have one of the forms $5x \pm 1$ or $5x \pm 2$. If $p = 5x \pm 1$, then either $p+1$ or $p-1$ is a multiple of 5, and hence so is $p^4 - 1$. If $p = 5x \pm 2$, then $p^2 + 1 = 25x^2 \pm 20x + 5$ is a multiple of 5, and again, so is $p^4 - 1$. So far we have shown that $p^4 - 1$ is divisible by $2^4 \cdot 3 \cdot 5 = 240$ for every prime $p > 5$. To see that these are the highest powers of 2, 3, 5 that can be used, consider $11^4 - 1 = 2^4 \cdot 3 \cdot 5 \cdot 61$. Next, suppose that some prime $P > 5$ divides $p^4 - 1$ for every prime $p > 5$. In particular, this would mean that P divides $P^4 - 1$. Since P already divides P^4 , this is not possible. The final answer is therefore **240**.